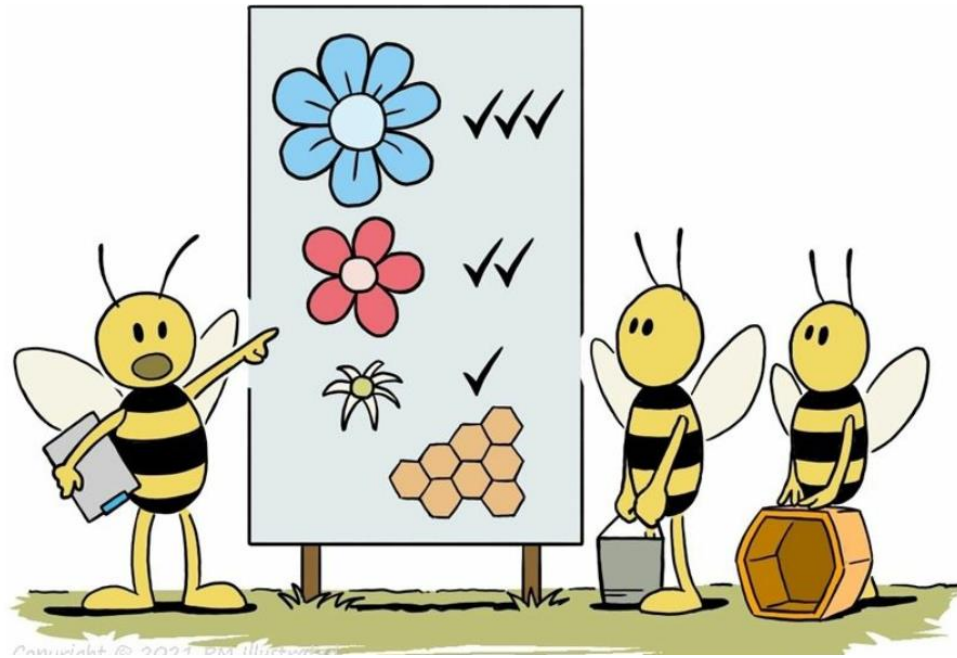


Project Management

5. Scope Management

Contents

- Scope Management
 - What is scope
- Plan Scope Management
- Collect Requirements
 - What is requirement?
- Define Scope
- Create WBS
 - How to decompose work
- Validate Scope
- Control Scope

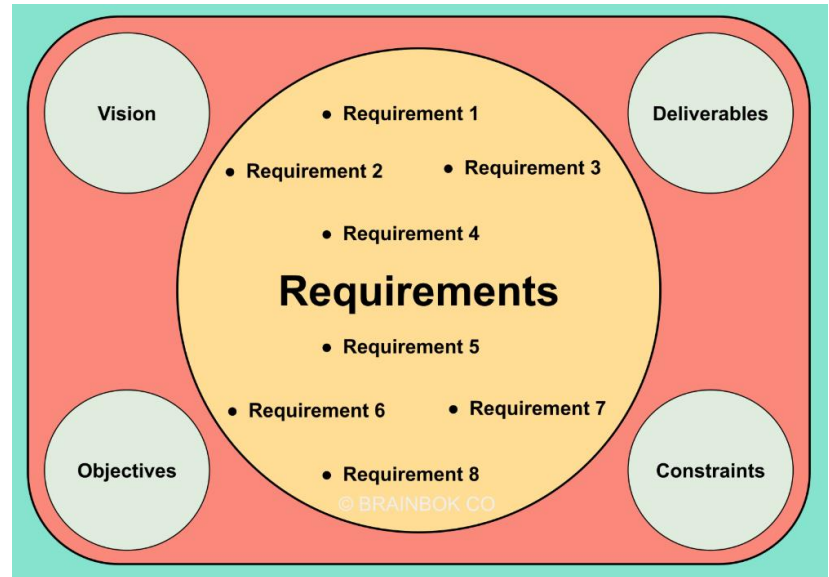


Scope Management

Understand the scope of what we are trying to do, and keep everyone on track.

What is scope

- Product scope means the *features* and *functions* of the product or service that you and your team are building.
- Project scope is *all of the work that needs to be done* to make the product.



Common mistakes

Scope Creep

- is also known as requirement creep, which refers to the uncontrolled changes (unofficial, gradual) in the project's or product's scope.



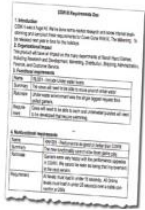
Gold Plating

- Adding extra features or functions to the products which were not asked by the customer.





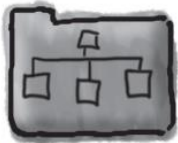
Project Management plan



Requirements documentation



Project scope statement



Work breakdown structure



Change requests



Accepted deliverables

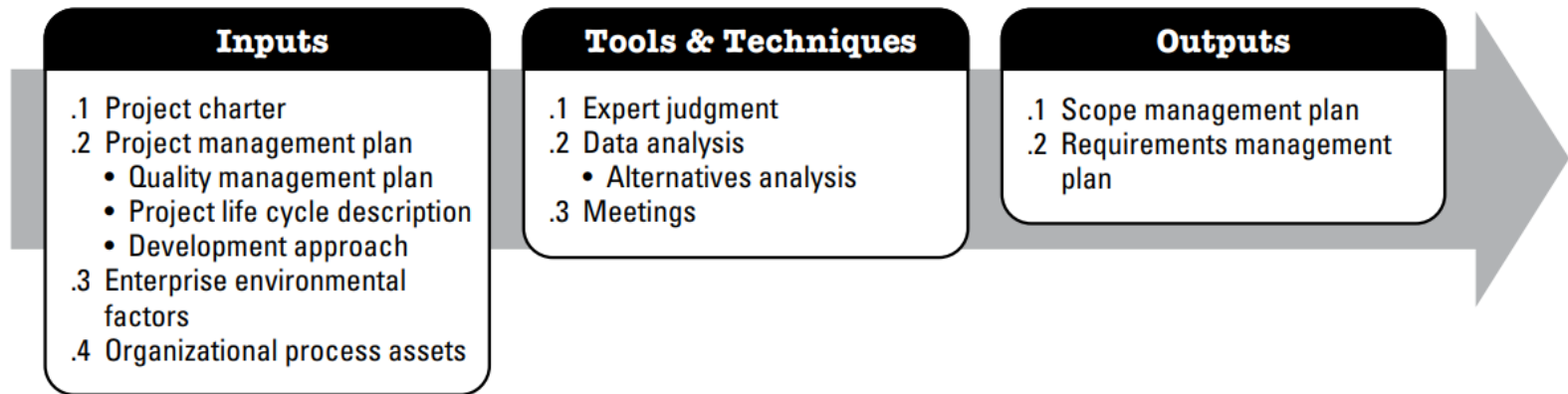
- **Plan Scope Management:** plan out all of the work you'll do to define your scope
- **Collect Requirements:** find out all of the stakeholders' needs
- **Define Scope:** write down a detailed description of the work you'll do and what you'll produce
- **Create WBS:** The work breakdown structure (WBS) organizes all of your team's work into work packages—or discrete pieces of work that team members do—so that you can keep the momentum of the project going from the start.
- **Control Scope:** when scope changes aren't controlled, it leads to the most frustrating sort of project problems.
- **Validate Scope:** make sure that what you're delivering matches what you wrote down in the project scope statement.

Plan Scope Management

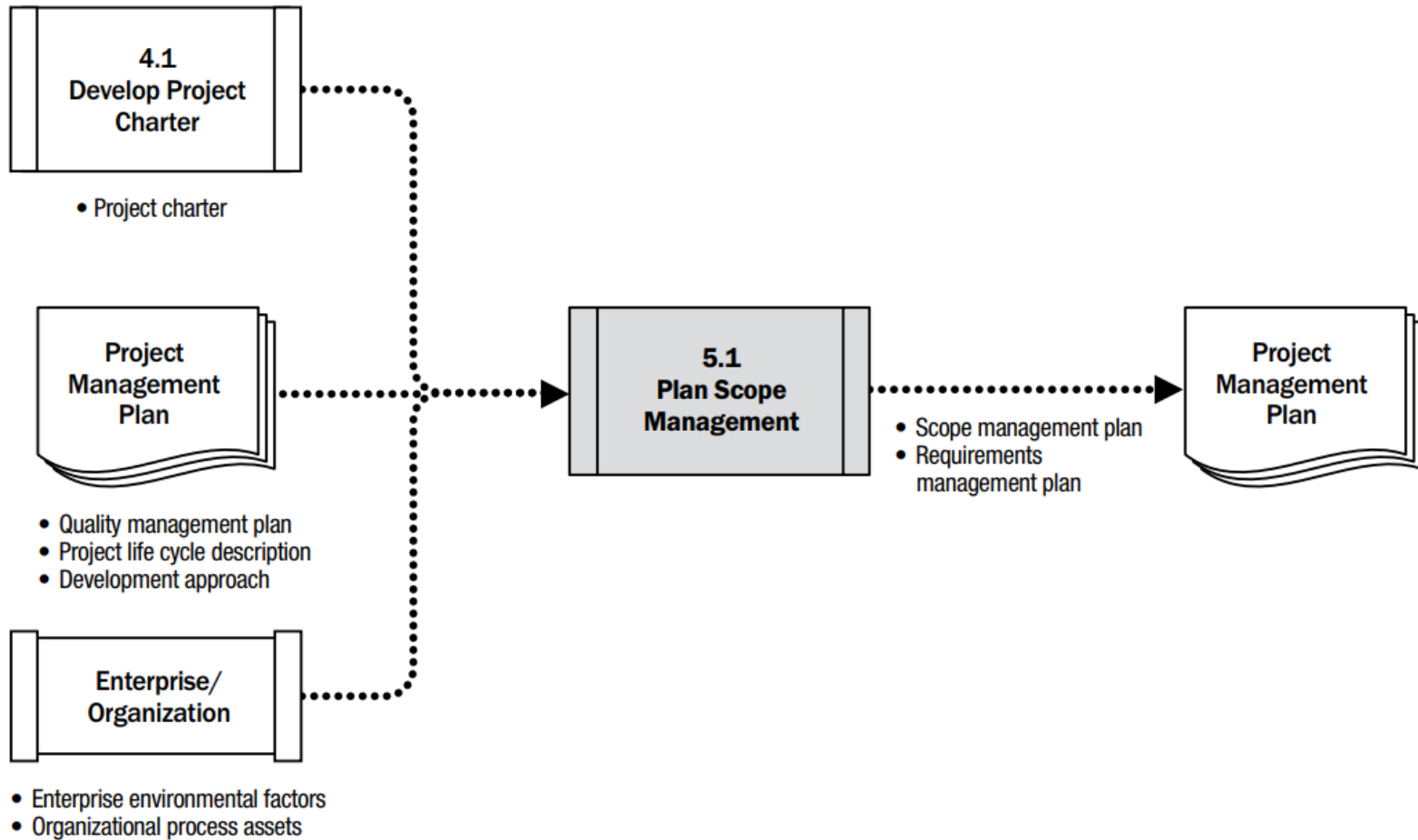
Plan Scope Management

- **What:** documents how the project and product scope will be defined, validated, and controlled.
- **Why:** provides guidance and direction on how scope will be managed throughout the project.
- **When:** performed once or at predefined points in the project.
- **How?**
 - Analysis of project charter, the latest approved subsidiary plans of the project management plan, historical information, and any other relevant enterprise environmental factors.
 - Determine way of collecting requirements, elaborating the project and product scope, creating the product, validating the scope, and controlling the scope are evaluated.

Process



Data Flow



Inputs

Project charter

- Project purpose, high-level project description, assumptions, constraints, and high-level requirements

Project management plan

- **Project life cycle description.**
- **Development approach.** Waterfall, iterative, adaptive, agile, or a hybrid development approach will be used.
- **Quality management plan.** Quality policy, methodologies, and standards are implemented on the project.

Enterprise environmental factors

- Organization's culture,
- Infrastructure, personnel administration, and marketplace conditions.

Organizational process assets

- Policies and procedures, and
- Historical information and lessons learned repositories.

Tools & Techniques

Expert Judgement

Expertise in:

- Previous similar projects
- Information in the industry, discipline, and application area

Meeting

- Attendees may include the project manager, the project sponsor, selected project team members, selected stakeholders

Alternatives analysis

- Evaluate the different ways to manage project requirement and project scope

Characteristics				
Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Manage cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent smaller deliveries	Speed
Agile	Dynamic	Repeated until correct	Frequent small deliveries	Customer value via frequent deliveries and feedback

Outputs: Scope Management Plan

- The scope management plan is a component of the project or program management plan that describes how the scope will be defined, developed, monitored, controlled, and validated.
 - Process for preparing a project scope statement;
 - Process that enables the creation of the WBS from the detailed project scope statement;
 - Process that establishes how the scope baseline will be approved and maintained;
 - Process that specifies how formal acceptance of the completed project deliverables will be obtained
- Can be formal or informal, broadly framed or highly detailed, based on the needs of the project.

SCOPE MANAGEMENT PLAN	
Project Title: _____	Date: _____
Work Breakdown (WBS) Structure	
WBS Dictionary	
Scope Baseline Maintenance	
Deliverable Acceptance	
Scope and Requirements Integration	
Page 1 of 1	

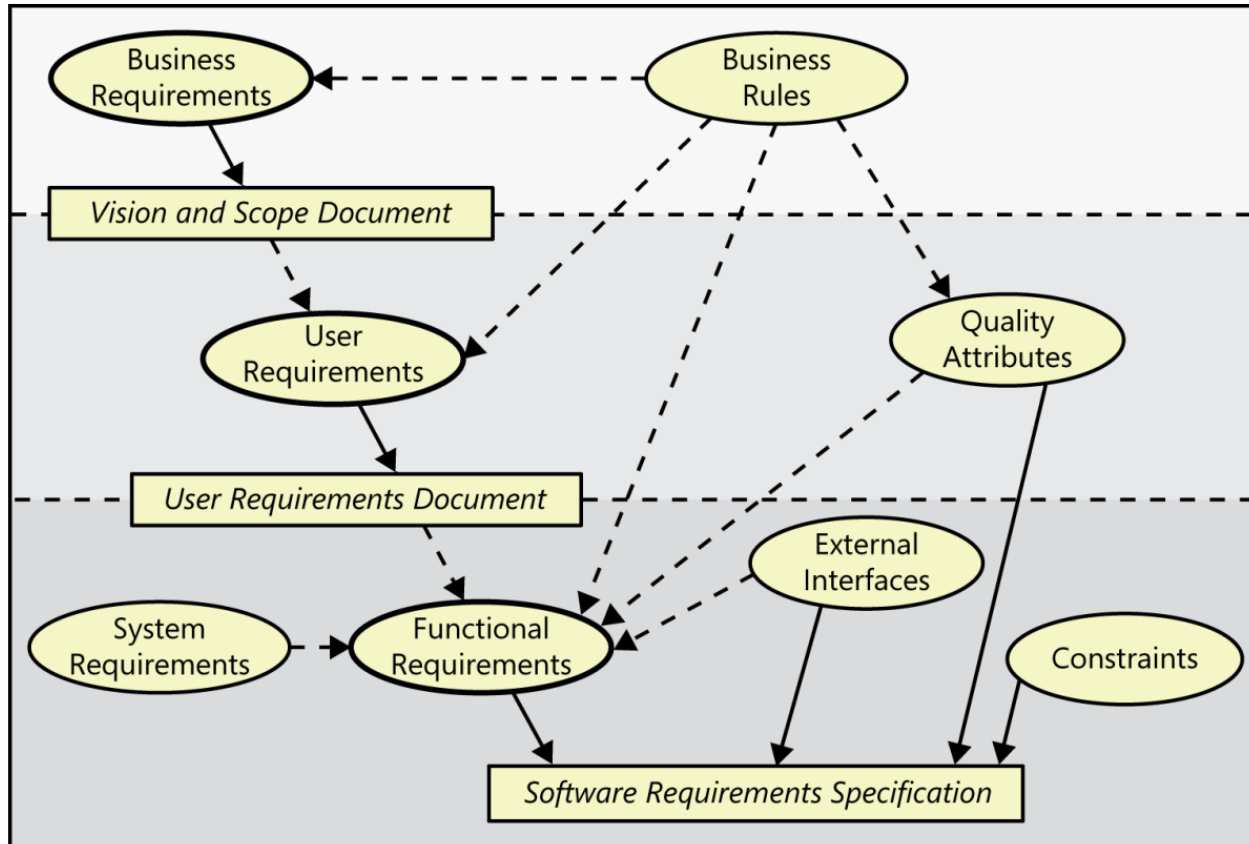
Outputs: Requirement Management Plan

- Describes how project and product requirements will be analyzed, documented, and managed.
 - How requirements activities will be planned, tracked, and reported;
 - Configuration management activities such as: how changes will be initiated; how impacts will be analyzed; how they will be traced, tracked, and reported; as well as the authorization levels required to approve these changes;
 - Requirements prioritization process;
 - Metrics that will be used and the rationale for using them;
 - Traceability structure that reflects the requirement attributes captured on the traceability matrix.
- Requirements management plan components are strongly influenced by phase-to-phase relationship

The image shows a template for a Requirements Management Plan. It is a vertical document with a light gray background and a white border. At the top, the title "REQUIREMENTS MANAGEMENT PLAN" is centered in a bold, black, sans-serif font. Below the title, there are two lines for "Project Title:" and "Date:". Following these are five sections, each with a label and a large rectangular box for content: "Requirements Collection:", "Categories:", "Prioritization:", "Traceability:", and "Configuration Management:". The final section is "Verification:", which is partially visible at the bottom. The document has a slightly aged, hand-drawn appearance with some irregular lines and a small tear at the bottom right corner.

Collect Requirements

What is requirement?



A **specific need or condition** that a product, service, or result must meet to satisfy stakeholders or achieve objectives.

Requirements vs Scope

Aspect	Requirements	Scope
Nature	Describes what is needed or expected.	Describes what will be delivered and what won't .
Purpose	To ensure stakeholder needs are understood and addressed.	To define the work boundaries and prevent scope creep .
Examples	"The app must support English and Spanish." "The system should process 1000 requests/sec."	"Develop a multilingual mobile app with login, profile, and dashboard modules."
Level	Can be granular and detailed.	More high-level and summarized.
When used	During requirements elicitation and validation.	During scope definition and planning.

Term	Definition
Business requirement	A high-level business objective of the organization that builds a product or of a customer who procures it.
Business rule	A policy, guideline, standard, or regulation that defines or constrains some aspect of the business. Not a software requirement in itself, but the origin of several types of software requirements.
Constraint	A restriction that is imposed on the choices available to the developer for the design and construction of a product.
External interface requirement	A description of a connection between a software system and a user, another software system, or a hardware device.
Feature	One or more logically related system capabilities that provide value to a user and are described by a set of functional requirements .
Functional requirement	A description of a behavior that a system will exhibit under specific conditions.
Nonfunctional requirement	A description of a property or characteristic that a system must exhibit or a constraint that it must respect.
Quality attribute	A kind of nonfunctional requirement that describes a service or performance characteristic of a product.
System requirement	A top-level requirement for a product that contains multiple subsystems, which could be all software or software and hardware.
User requirement	A goal or task that specific classes of users must be able to perform with a system, or a desired product attribute.

Collect Requirements

- **What:** process of determining, documenting, and managing stakeholder needs and requirements to meet objectives
- **Why:** provides the basis for defining the product scope and project scope.
- **When:** performed once or at predefined points in the project.

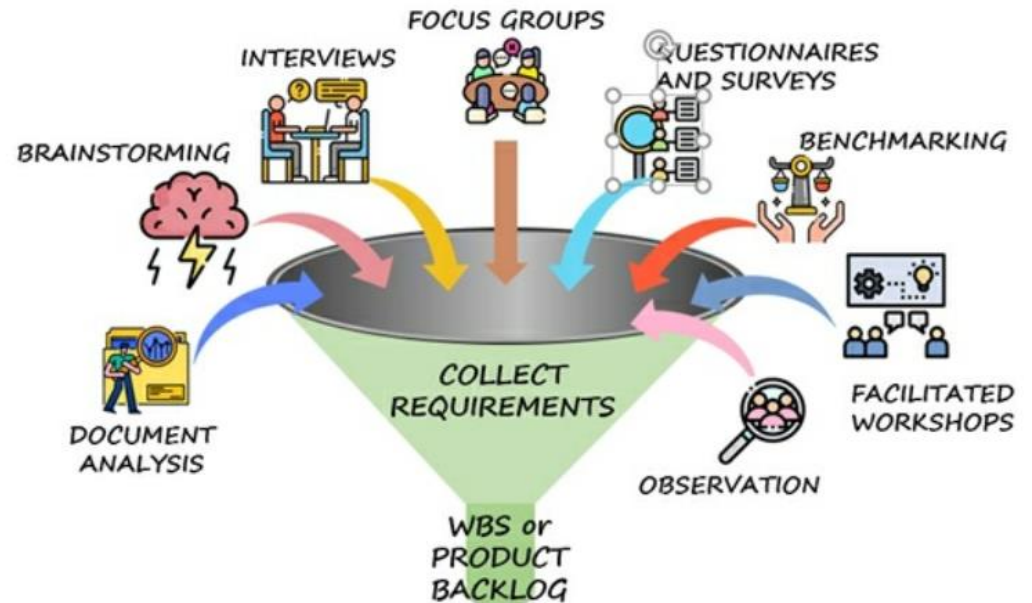
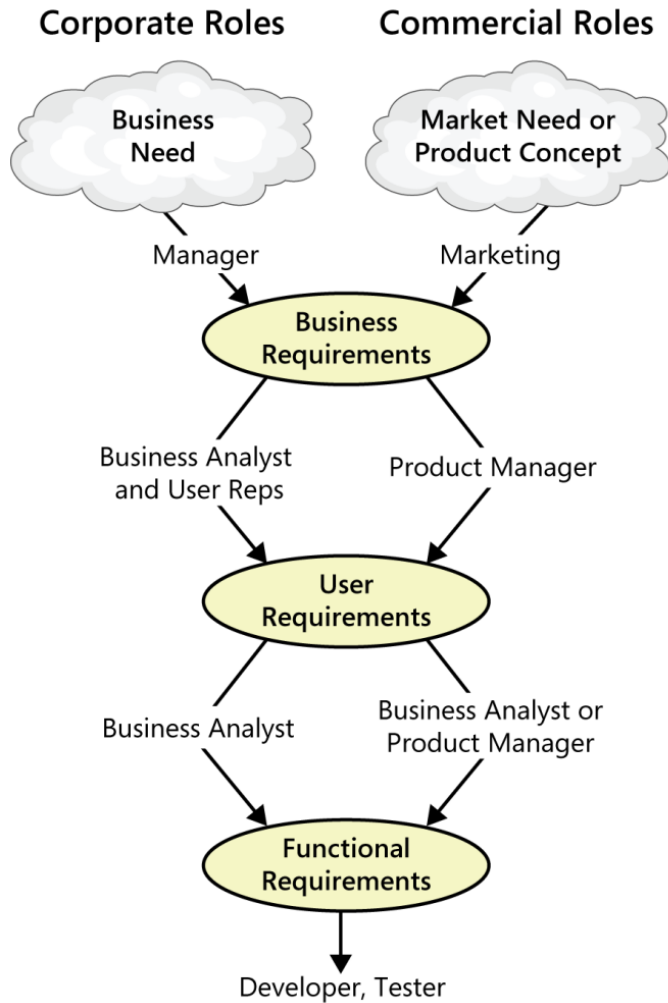
How stakeholders think requirement gathering works.



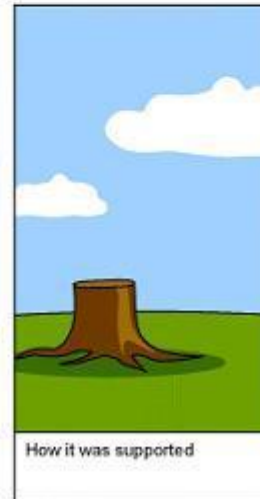
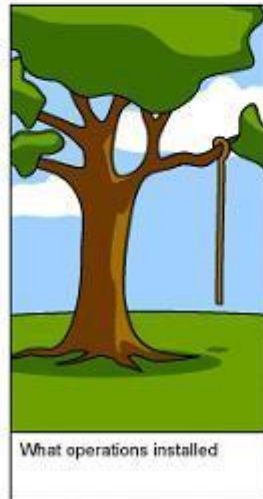
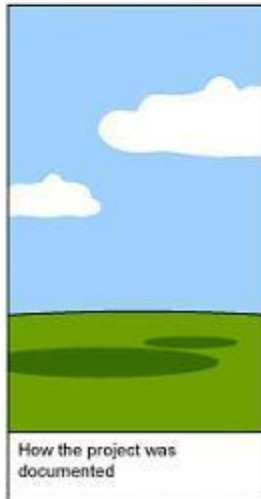
How requirement gathering really works.



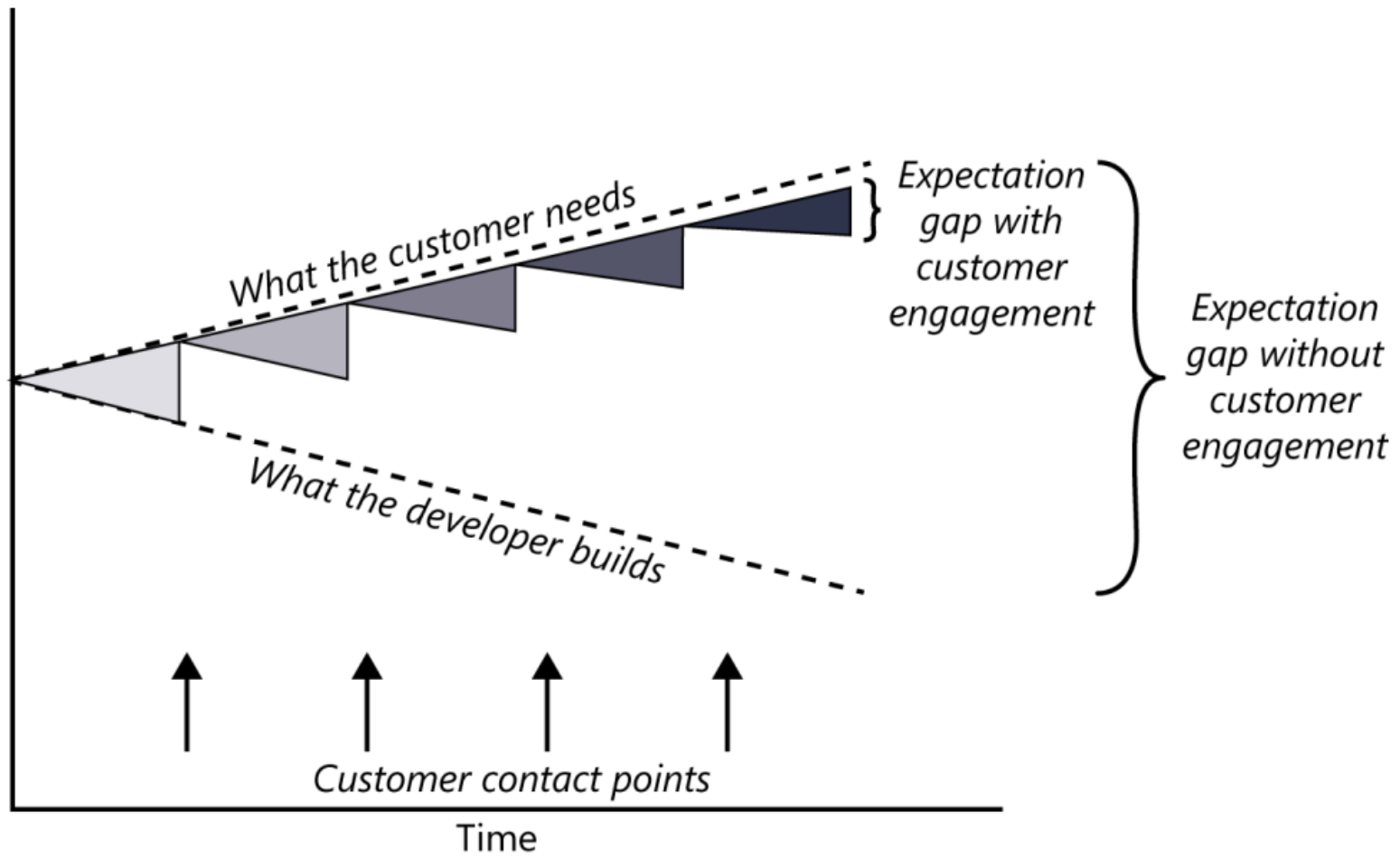
Collect requirements



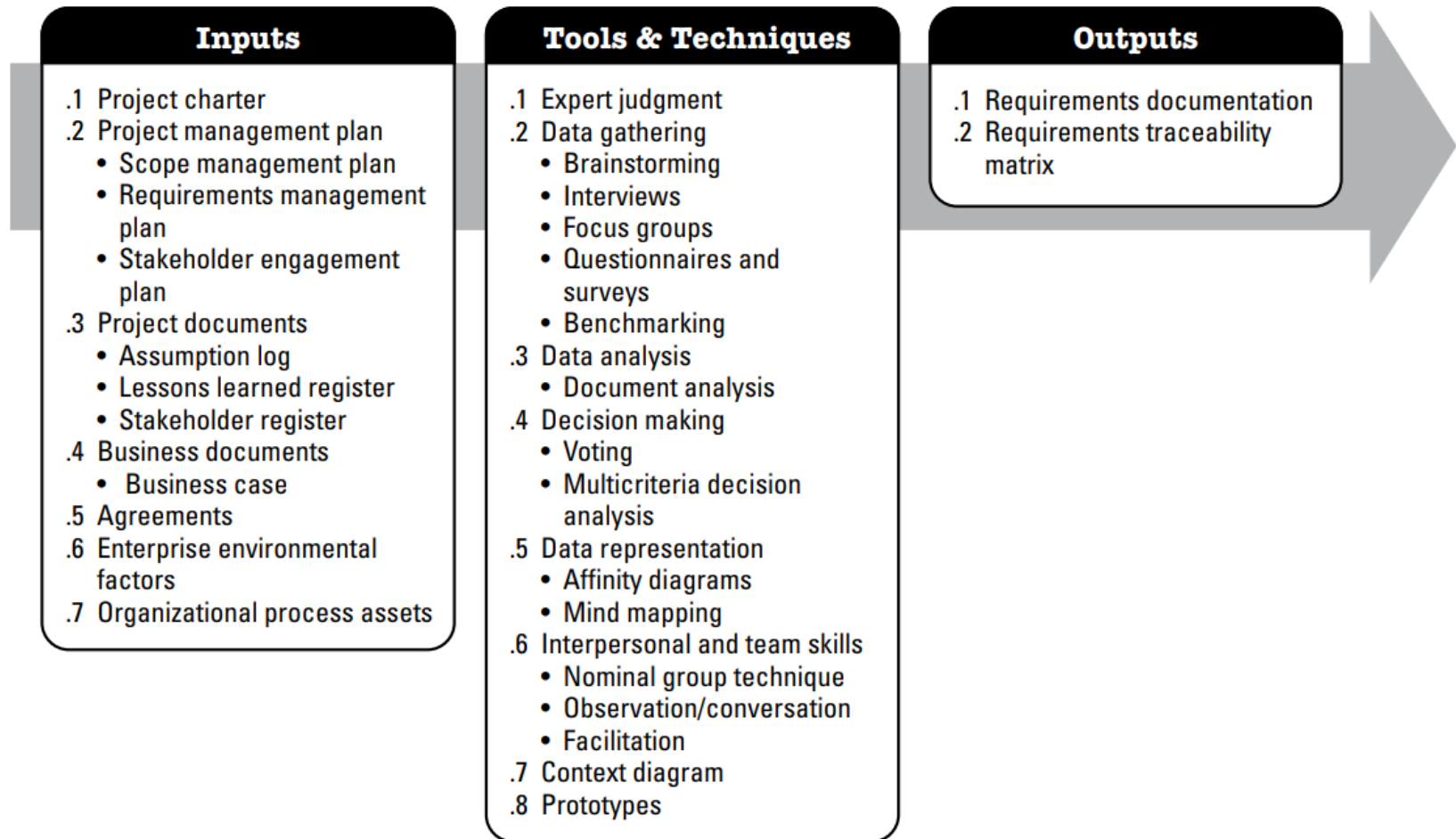
How it really work



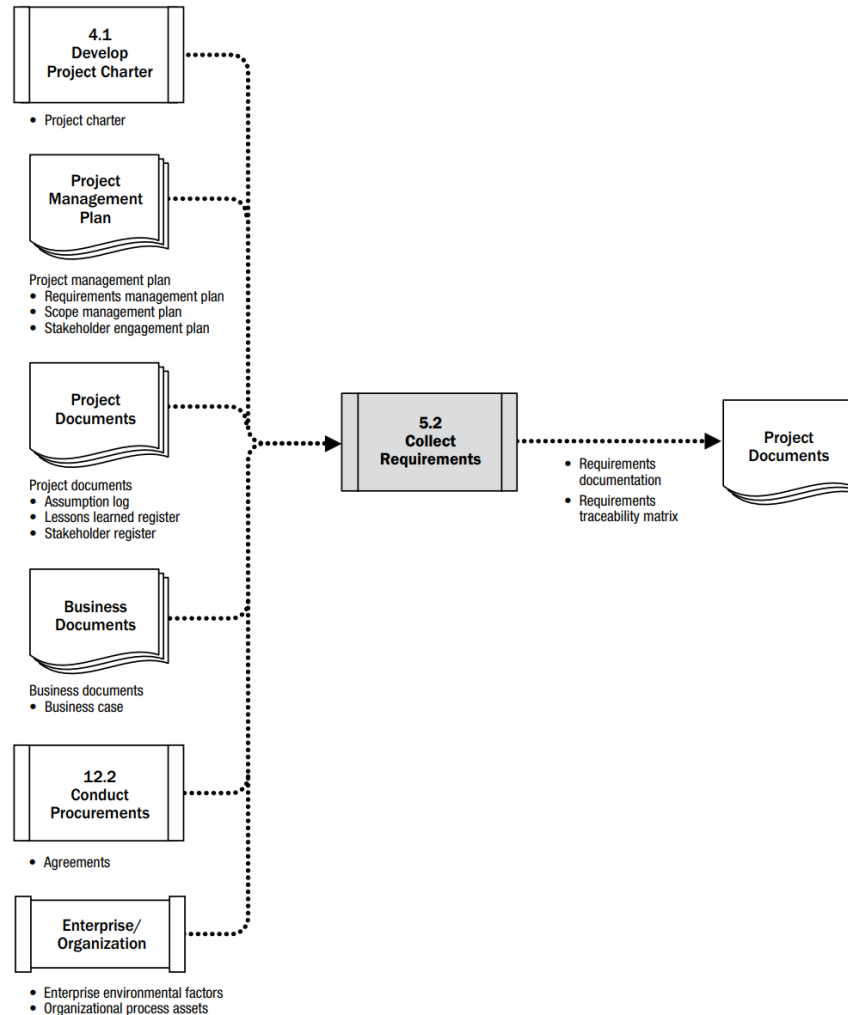
The expectation gap



Process



Data Flow



Requirements development

Step-by-Step Guide to Creating Software Development Requirements



? Elicitation

Discussions with stakeholders, reviewing existing documentation, and **understanding the project's goals and environment**. The aim is to gather information about the software system's **use cases and quality attributes**.

Analysis

Examining and refining the requirements gathered during elicitation. This may involve **grouping, eliminating, and prioritizing** requirements based on **stakeholder needs** and goals.

Specification

Documenting software development requirements in an SRS document. The SRS should **describe each requirement clearly** and **guide the development process**.

Validation

Verifying that the software development requirements are **realistic, valid, unambiguous, complete, and demonstrable**. This may involve using prototypes, use cases, reviews, or tests.

Management

Documenting, tracing, and agreeing on the software development requirements throughout the process. It involves **maintaining and updating the requirements** as the software development progresses and communicating any changes to stakeholders.

Tools & Techniques: Data analysis

Document analysis

- There are a wide range of documents that may be analyzed to help elicit relevant requirements.
- **Example:** *business plans, marketing literature, agreements, requests for proposal, current process flows, logical data models, business rules ...*



Tools & Techniques: Data-gathering

Brainstorming:

- Gathering a list of ideas spontaneously contributed by its members.
- Rules for brainstorming :
 - Go for **quantity**
 - Withhold **criticism**
 - Welcome wild ideas
 - Combine and improve ideas



Interviews

- Is a formal or informal approach to discover information from stakeholders by talking to them directly
- **Individually meeting** between an Interviewer and an interviewee

Focus Groups

- **Group of prequalified stakeholders** or the subject matter experts
- A trained moderator guides the group through an interactive conversational than a discussion , designed to be more conversational than a one-on-one interview

Tools & Techniques: Data-gathering

Questionnaires and Surveys

- Questionnaires and surveys are written sets of questions designed to quickly accumulate information from a wide number of respondents .



Benchmarking

- Generate ideas for improvement, and provide a basis for measuring performance by comparing actual or planned practices, such as processes and operations, to those of comparable organizations to identify best practices

Competitor Benchmarking

	You	Competitor 1	Competitor 2	Competitor 3
Product quality	9	1	8	5
Pricing	3	4	3	2
Place	3	7	4	8
Promotion	4	7	9	4
Positioning	6	9	7	10
Reputation	4	10	4	8
People	7	3	7	2
Partnership	8	7	8	6

Tools & Techniques: Decision-making

Voting

- **Unanimity:** A decision that is reached whereby everyone agrees on a single course of action.
- **Majority:** A decision that is reached with support obtained from more than 50% of the members of the group.
- **Plurality:** A decision that is reached whereby the largest block in a group decides, even if a majority is not achieved.



Autocratic

One individual makes the decision for the group



Multi-criteria decision analysis

Uses a **decision matrix** to provide a systematic analytical approach for establishing criteria, such as risk levels, uncertainty, and valuation, to evaluate and rank many ideas.

Decision matrix

- A Decision Matrix (also known as a criteria matrix, Pugh matrix, or weighted scoring model) is a structured tool used to evaluate and prioritize multiple options based on a set of weighted criteria.
 - It helps teams make objective decisions by scoring each option against predefined criteria, rather than relying on opinions or intuition.
- A decision matrix helps:
 - Rank and prioritize option.
 - Choose between multiple solution options.
 - Justify decisions with a transparent, consistent process
- When to use a Decision Matrix in requirement collection
 - Prioritizing features: Choosing MVP scope or sprint backlog items
 - Selecting tools or vendors: Picking between Jira, Trello, or ClickUp
 - Choosing implementation strategies: API-first vs monolithic vs microservices
 - Aligning stakeholders: Making transparent, rational group decision

Options:

1. User Profile
2. Push Notifications
3. In-app Chat

Criteria:

- Business Value (Weight = 5)
- Customer Demand (Weight = 4)
- Development Effort (Weight = 3, lower is better)
- Technical Risk (Weight = 2, lower is better)

*For effort and risk, higher score = worse
→ can reverse score or invert weights as needed.*

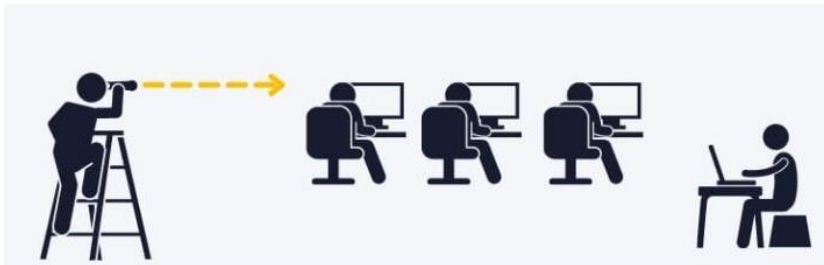
Feature	Biz Value (5)	Demand (4)	Dev Effort (3)	Risk (2)	Total
User Profile	$5 \times 5 = 25$	$4 \times 4 = 16$	$2 \times 3 = 6$	$1 \times 2 = 2$	49
Push Notification	$3 \times 5 = 15$	$5 \times 4 = 20$	$1 \times 3 = 3$	$2 \times 2 = 4$	42
In-app Chat	$4 \times 5 = 20$	$3 \times 4 = 12$	$5 \times 3 = 15$	$4 \times 2 = 8$	55

→ Chosen feature: User Profile (highest score = 49)

Tools & Techniques: interpersonal and team skills

Observation/conversation

- Observation (also called “**job shadowing**”) is usually done externally by the observer viewing the user performing his or her job .
- It can also be done by a “**participant observer**” who actually performs a process to experience how it is done to uncover hidden requirements



Nominal Group Technique:

- Brainstorming + Voting

Voting by Team Member (8 points for 1st choice, 7 for 2nd, 6 for 3rd, ...)					
Solution Options	Ray	Robin	Fred	Points	Ranking
EZTrain	4	5	4	13	-
WebCast	8	7	8	23	1st
B2BLearn	6	3	5	14	-
BlendNow	1	1	2	4	-
LearnFast	3	4	3	10	-
CBTRun	7	8	6	21	2nd
NowClick	2	2	1	5	-
MyTrainer	5	6	7	18	3rd

Tools & Techniques: interpersonal and team skills

Facilitated workshops

- can be used to quickly define **cross – functional** requirements and reconcile stakeholder differences.

Examples of facilitated workshops:

- **User stories**: short, textual descriptions of required functionality, are often developed during a requirements workshop
- **Joint application design/development (JAD)**: bringing business subject matter experts and the development team together to gather requirements and improve the software development process.

Facilitation technique

- Well-facilitated sessions can build trust, foster relationships, and improve communication among the participants, which can lead to increased stakeholder consensus.



Tools & Techniques

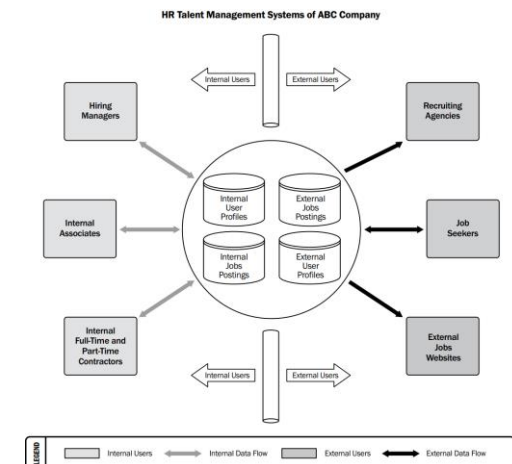
Prototypes

- Concept, Mock up, Storyboard, Demo...a working model of the expected product before actually building it
- The requirements obtained from the prototype are sufficiently complete to move to a design or a build phase



Context diagrams

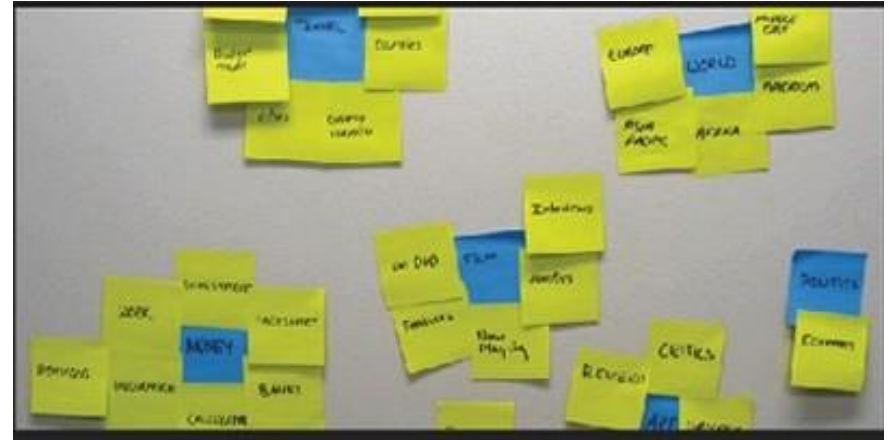
- A diagram that represents the Actors outside a system that could interact with that system.
- The actor(s) providing the input to and receiving the output from business system.
- This diagram is the highest level view of a system



Tools & Techniques: Data representation

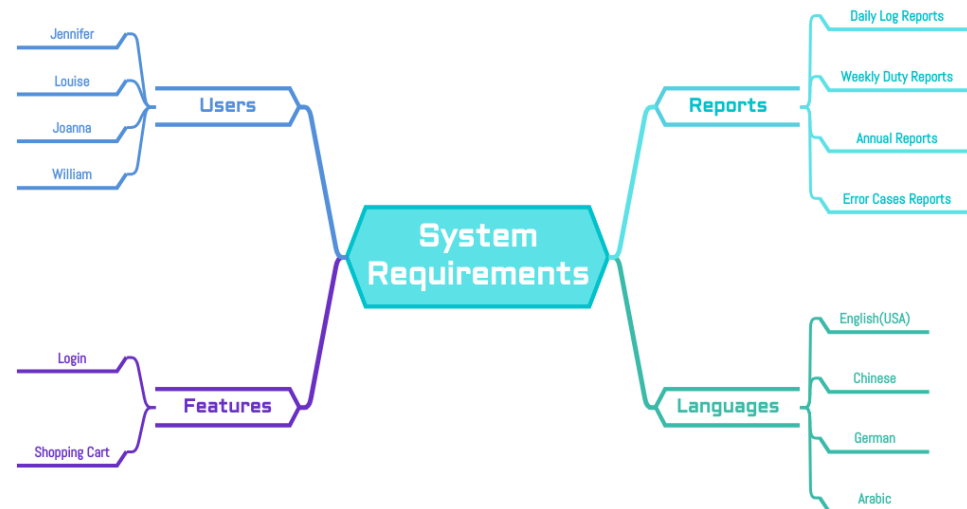
Affinity Diagram:

- Brainstorming+ Grouping



Mind Map:

- Brainstorming + Mapping



Requirements Traceability Matrix

- | REQUIREMENTS TRACEABILITY MATRIX | | | | | | | | |
|---|-------------|--------|----------|----------|---------------------------|-------------|--------------|------------|
| Project Title: _____ | | | | | Date Prepared: _____ | | | |
| Requirement Information | | | | | Relationship Traceability | | | |
| ID | Requirement | Source | Priority | Category | Business Objective | Deliverable | Verification | Validation |
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Page 1 of 2

Requirements Traceability Matrix

Req ID	Requirement Description	Business Objective	Design Doc	Code Module	Test Case	Status
R-001	User can log in using username & password	Ensure secure user access	Login_UI.md	authController.js	TC-01	✅ Passed
R-002	Display account balance after login	Provide users with quick access to info	Dashboard_UI.md	balanceService.js	TC-05	✅ Passed
R-003	Users must reset password after 3 failed attempts	Enhance security	Password_Policy.doc	securityManager.java	TC-09	❌ Failed
R-004	Send OTP for transaction over \$1000	Fraud prevention	OTP_Flow.vsd	otpServices	TC-11	🟡 In Progress

✅ Traceability

✅ Test Coverage

✅ Change Impact

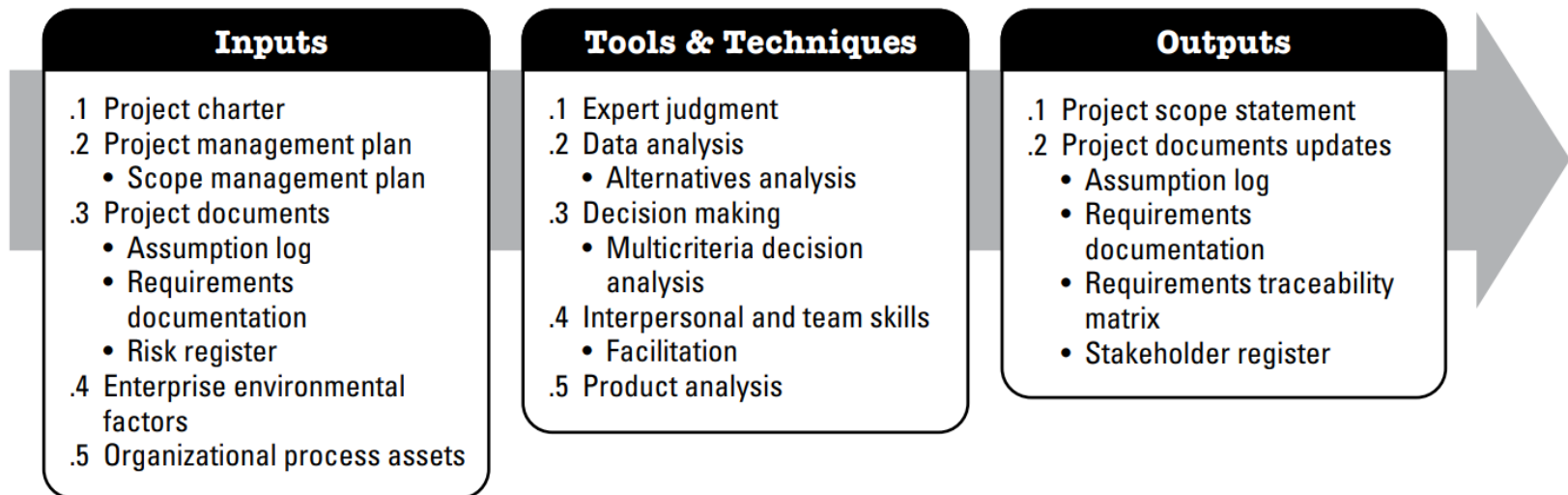
✅ Audit-Ready

Define Scope

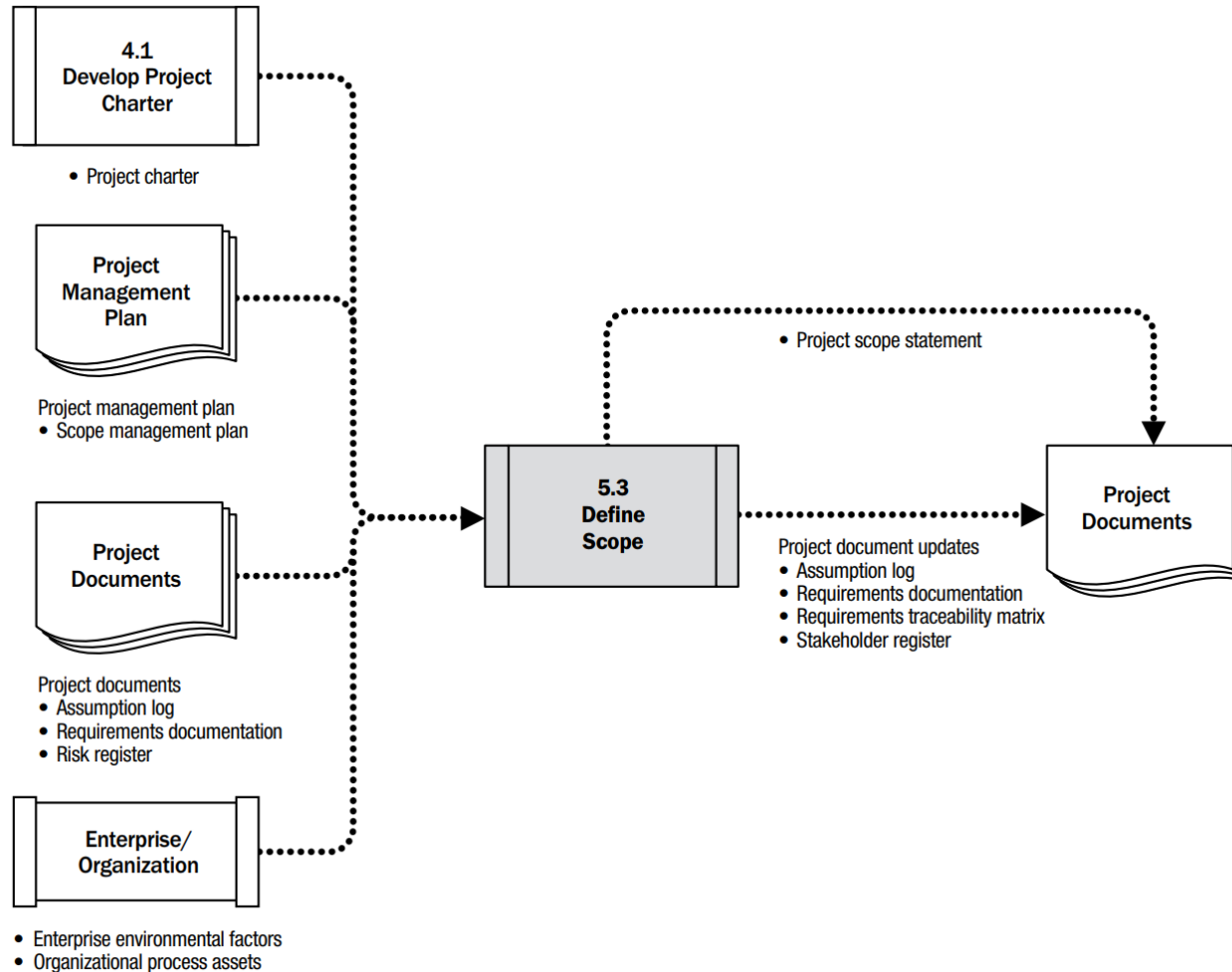
Define Scope

- **What:** process of developing a detailed description of the project and product scope.
- **Why:** describes the product, service, or result boundaries and acceptance criteria.
 - The requirements identified in Collect Requirements may not be included in the project
- **When:** Depends on project lifecycle, Define scope process can be used once or highly iterative
- **How:**
 - Builds upon the major deliverables, assumptions, and constraints that are documented during project initiation
 - During project planning, the project scope is defined and described with greater specificity as more information about the project is known.
 - Existing risks, assumptions, and constraints are analyzed for completeness and added or updated as necessary.
 - The process can be highly iterative.

Process



Data Flow



Tools and Techniques

Alternative analysis:

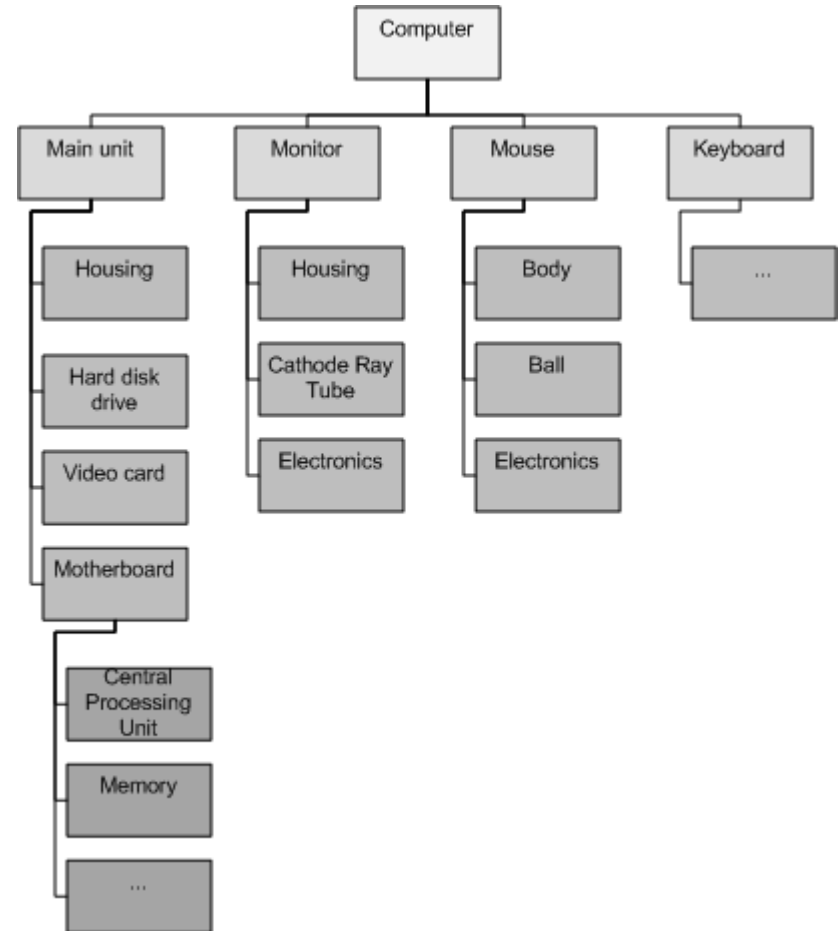
- Generate and analysis different approaches to execute and perform the work of the project.
- A variety of general management techniques can be used, such as:
 - *Brainstorming, Lateral thinking, Analysis of alternatives...*



Tools and Techniques

Product Analysis

- Methods for translating high-level product descriptions into tangible deliverables. Each application area has one or more generally accepted methods: *Product breakdown*, *Systems analysis*, *Requirements analysis*, *Systems engineering*...
- Generally asking questions about a product and forming answers to describe the use, characteristics, and other the relevant aspects of what is going to be manufactured.



Outputs

Project Scope Statement

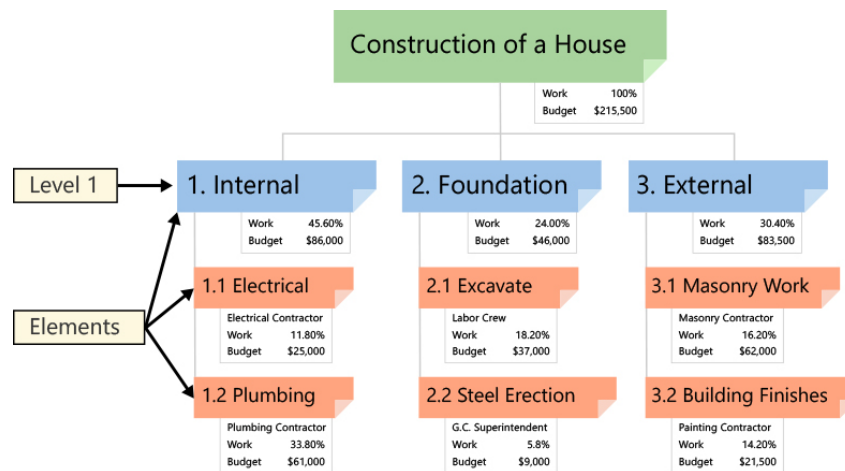
- Project scope statements documents the entire scope, including project and product scope. It describes the project's deliverables in detail.
- It helps to create a **common understanding** among stakeholders (avoid scope creep).
- It may contain explicit scope exclusions that can assist in managing stakeholder expectations.
- The project team and the project stakeholders need to agree to the scope statement before project execution.

Project Scope Statement			
Project Name	PATHWAYS PROJECT SCOPE STATEMENT		
Project Sponsor	Coda-edu	Project Manager	James Walsh
Date of Project Approval	02 March 2020	Last Revision Date	02 March 2020
Scope Description	Pathways will develop and implement course maps for all certificates and degrees with the objective of preventing students from taking courses that do not transfer or prepare them for a career. Further, we will analyze and design student support processes and areas that increase students' ability to successfully navigate their way through their educational experience.		
Project Deliverables	➤ Create Transfer Pathways maps for all viable degrees and Career Pathways maps for all viable certificates ➤ Create Student intake and support systems that have well-defined Pathways to student completion of educational goals ➤ Develop Student-centered classrooms with classroom content driven by Student Learning Outcomes (SLOs) and not textbook		
Acceptance Criteria	➤ Successful implementation of Degree/Certificate maps ➤ Successful implementation of Student Support maps ➤ Initial reduction in average credit hours to degree or certificate of 4% ➤ Initial increase in completion rates of 4%		
Project Exclusions	Since Student Success is the most important value of University of Leeds and all areas of the College have a direct or indirect affect on student success, all employees will be expected to support Pathways with their time and expertise. Some work will be occurring concurrently and will complement Pathways development, such as: ➤ SACSCOC ten-year reaffirmation of accreditation ➤ Open Educational Resources textbook review and selection		
Constraints	➤ Pathways work should follow the Values of University of Leeds ➤ Transfer/Career Pathways should follow THECB and other regulatory board requirements ➤ Pathways should be implemented by fall 2022 with review and necessary modifications in the 2022-2023 academic year ➤ Budget ➤ Time/resources/personnel limitations ➤ College Leadership will support the project and provide timely responses to requests ➤ Project staff will be dedicated to this project and will not be reassigned ➤ Project team members will be backfilled as necessary to support this project ➤ Funding for recommended and approved changes will be provided		

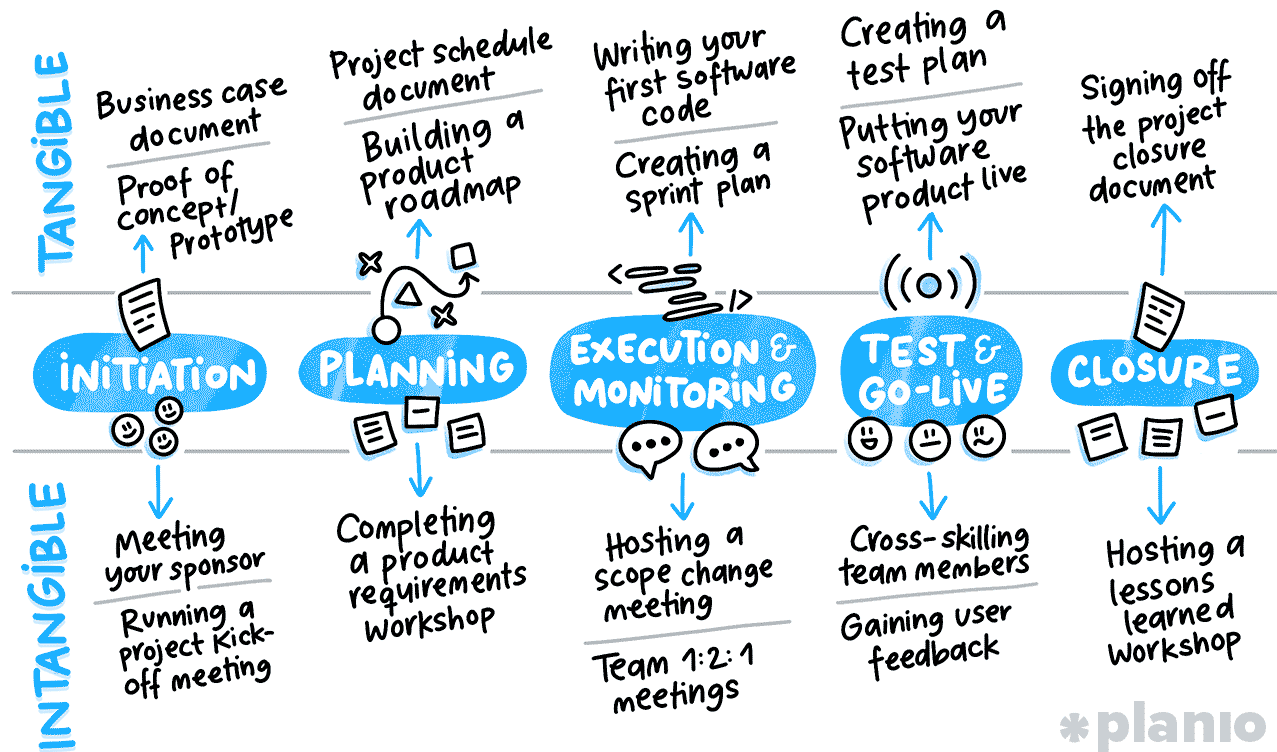
Create WBS

Create WBS

- **What:** process of subdividing project deliverables and project work into smaller, more manageable components.
- **Why:**
 - “Divide to conquer”
 - Easier to estimate, allocate resources and to track project performance
 - WBS visually defines the scope into manageable chunks that stakeholders can understand
- **When:** performed once or at predefined points in the project.

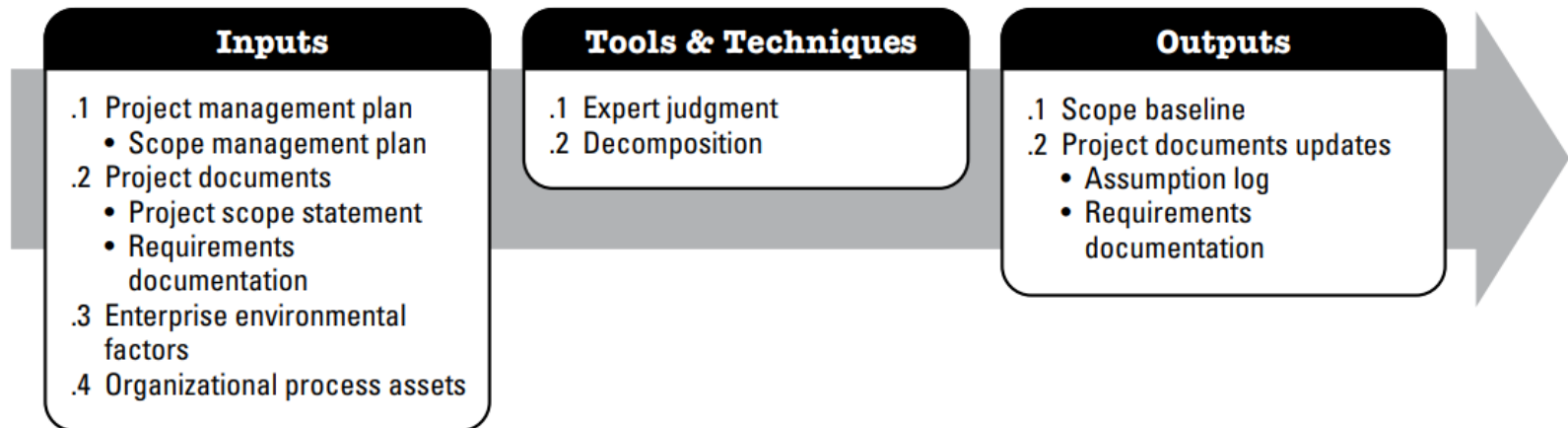


COMMON DELIVERABLES AT EACH STAGE OF A PROJECTS LIFECYCLE

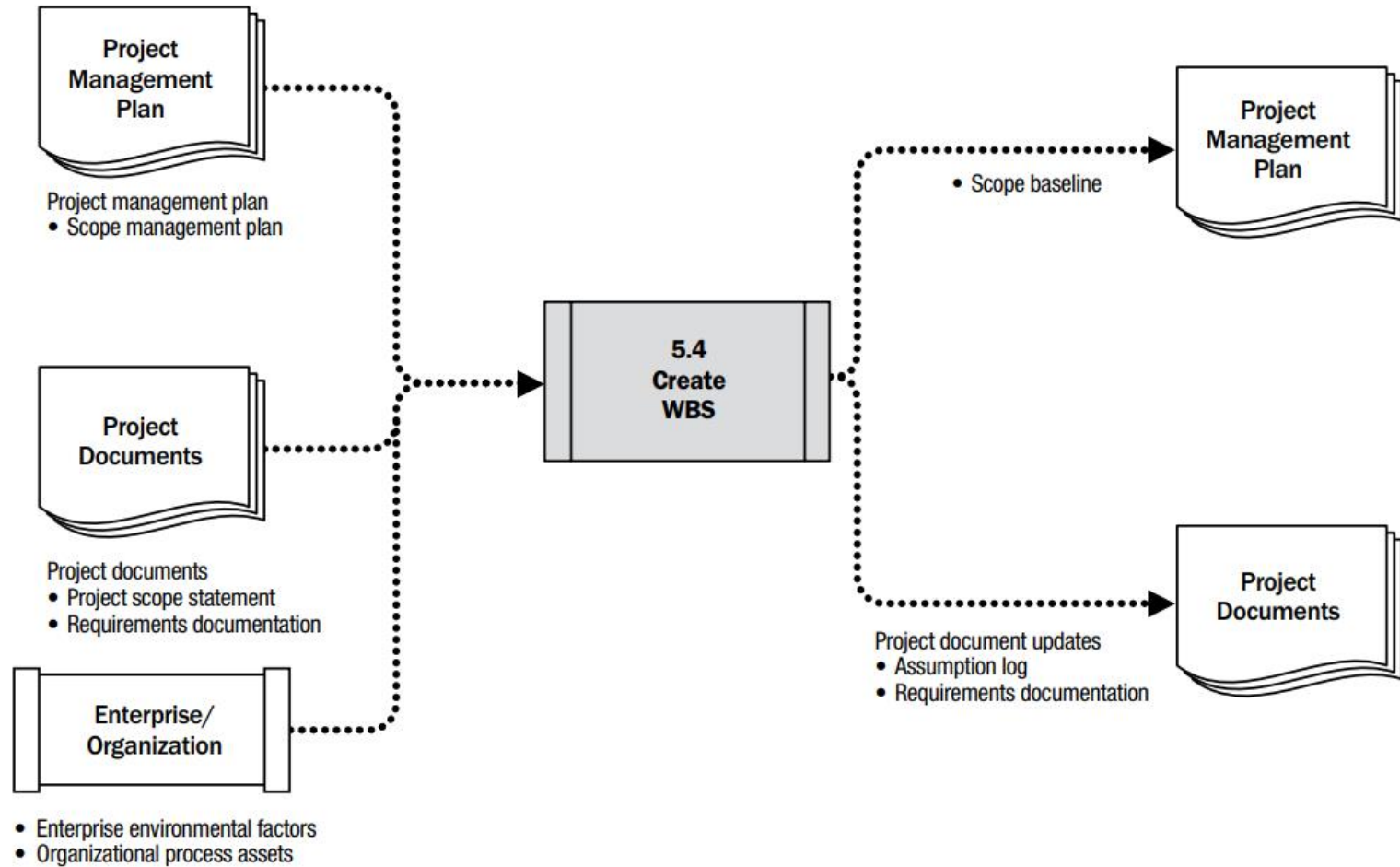


*Project **deliverables** are the specific, tangible and intangible **outputs** produced during a project that help it move towards its objective.*

Process

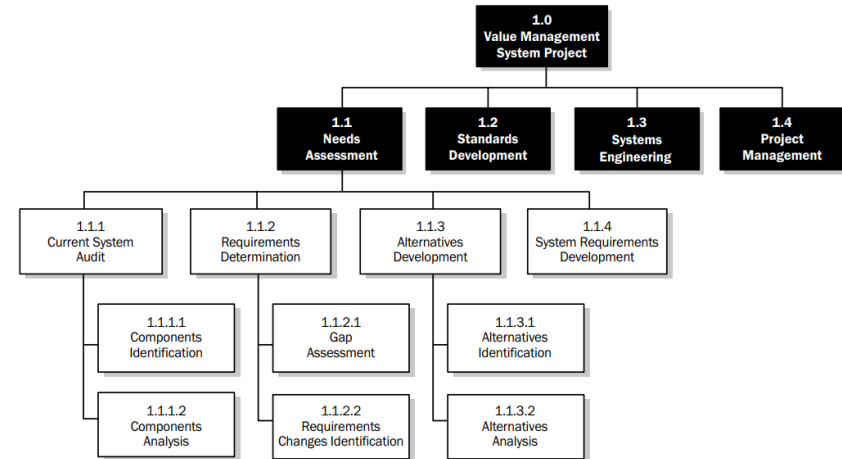


Data Flow



Tool & Techniques: Decomposition

- Identifying and analyzing the deliverables and related work
- Structuring and organizing the WBS
- **Decomposing** the upper WBS levels into lower-level detailed components
- Verifying that the degree of decomposition of the deliverables is appropriate.
- Developing and assigning identification codes to the WBS components
- Get approved and baseline project scope documents



When decomposition is impossible

Rolling wave planning: is a project planning technique that involves progressive elaboration to add detail to the Work Breakdown Structure (WBS) on an ongoing basis.

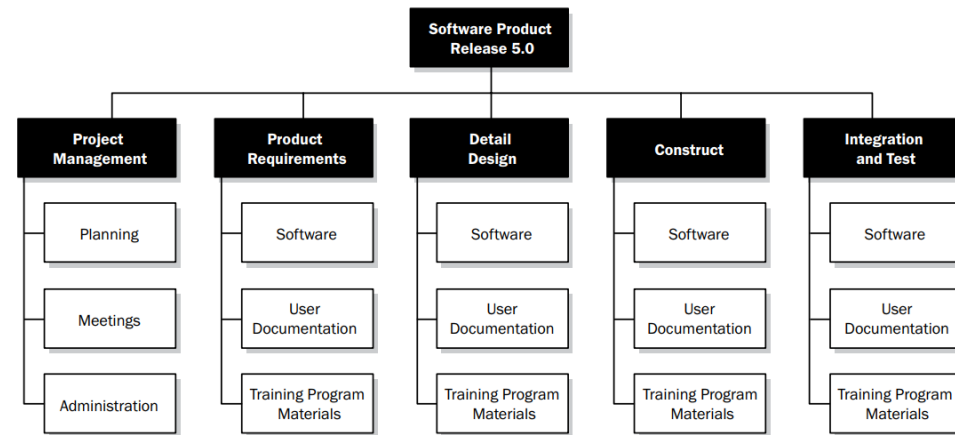
- **Near term deliverables:** are decomposed into activities necessary to produce the deliverables. Deliverables are also called as **Work packages**.
- **Long term deliverables:** are more broadly defined, called as **Planning packages**.

Work package	Planning package
• Lowest level element of WBS	• Lowest level element of WBS at a given point of time
• No further decomposition (in term of Scope management)	• Will get decomposed into work packages at later stage
• Primary input to identify activities process	• Decomposed into work package or they get converted into work package when work get more visibility • Planning package will not have activities under them

Tool & Techniques: Decomposition

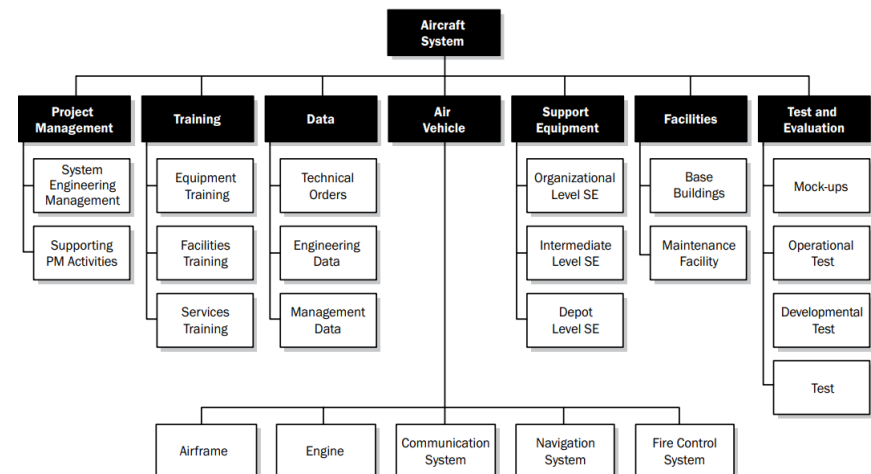
Approaches

- **Top-down:** use WBS templates or organization-specific guidelines.
- **Bottom-up:** constructed from the inputs of project team members who actually do the work (team buy-in)



Forms

- The technique involves breaking down the project into smaller, more manageable components of work.
- WBS can be organized by :
 - Project phases
 - Major deliverables and subprojects
 - Combination approach



How to decompose work?

- The lowest level of WBS components always are deliverables
- **Note: Plan deliverables (outcomes), not actions !**

What WBS is Not	What WBS is
• A WBS is not an exhaustive list of work.	• It is instead a comprehensive classification of project scope.
• A WBS is neither a project plan, a schedule, nor a chronological listing.	• It specifies what will be done, not how or when.
• A WBS is not an organizational hierarchy	• It may be used when assigning responsibilities.

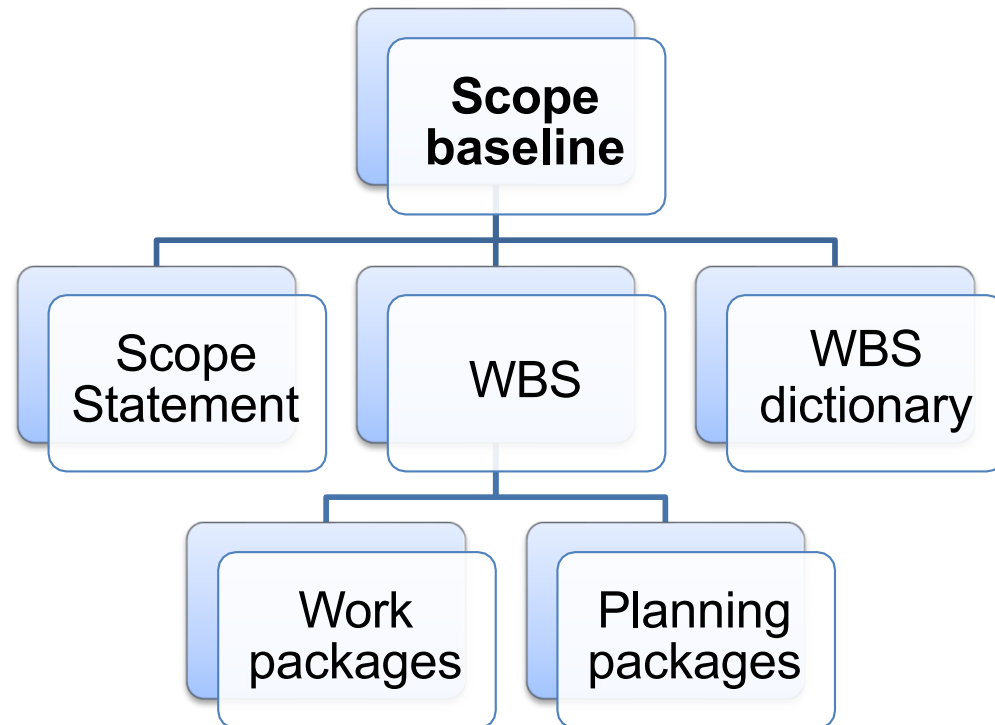
How to decompose work?

- **The level of decomposition** is often guided by the degree of control needed to effectively manage the project.
 - The level of detail for work packages will vary with the size and complexity of the project.
 - ***Excessive decomposition*** can lead to nonproductive, micro-management effort
 - Small enough: If you can assign it, measure it, and manage it ***without needing to break it further*** — it's small enough.
- Rule of thumb (Heuristic):
 - No workpackage should be more than a single reporting period
 - **The 8/80 Rule:** no work package should not be less than 8h and not be bigger than 80h of work
 - **Rule of 100%:** The sum of the work at the “child” level must equal 100% of the work represented by the “parent”, and the WBS should not include any work that falls outside the actual scope of the project

Outputs

Scope Baseline

- The approved version of a scope statement, WBS, and its associated WBS dictionary
- Scope baseline is a component of the project management plan.
- It can be changed only through formal change control procedures and is used as a basis for comparison.



Project Documents Updates

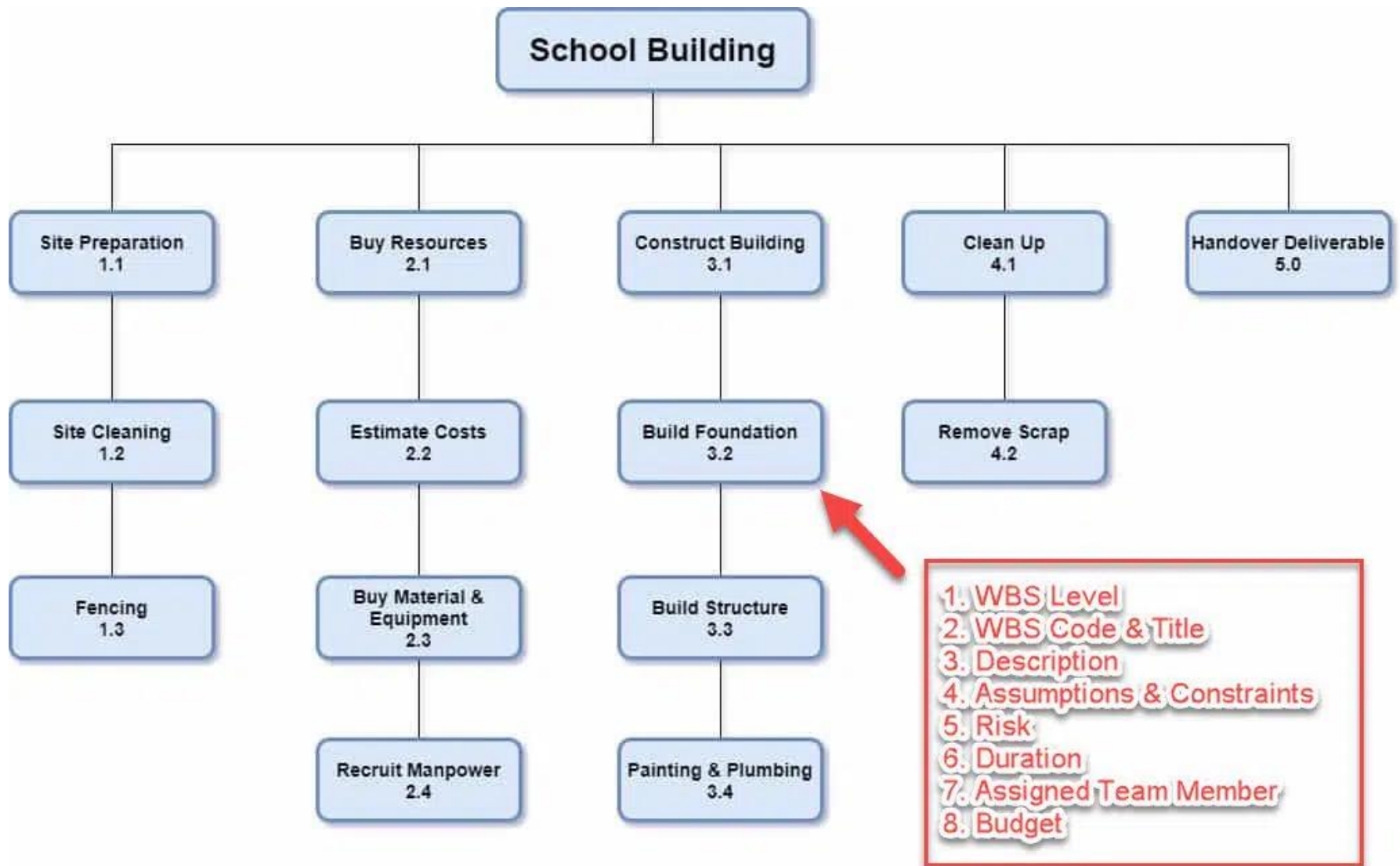
- **Assumption log.**
additional assumptions or constraints
- **Requirements documentation.**

Outputs

WBS dictionary

- Document that supports the WBS where detail work descriptions are documented.
- Information in the WBS dictionary may includes:
 - **Code of accounts:** a numbering system used to uniquely identify each component of the work breakdown structure (WBS).
 - Description of work
 - Assumptions and constraints
 - Responsible organization
 - Schedule milestones...

Project: Online Checkout System	
Work Package ID: 1.2.3.4	
Work Package Name: Design Checkout Funnel	
Work Package Description: Design the complete checkout panel, including payment screens, order confirmation page, etc.	
Assigned Team Member: Rob Breton	
Assigned Date: 10/22/2021	Date Due: 10/30/2021
Budget: 1,000 USD	Accounting Code : XYZ- 0450



Delivery Based WBS

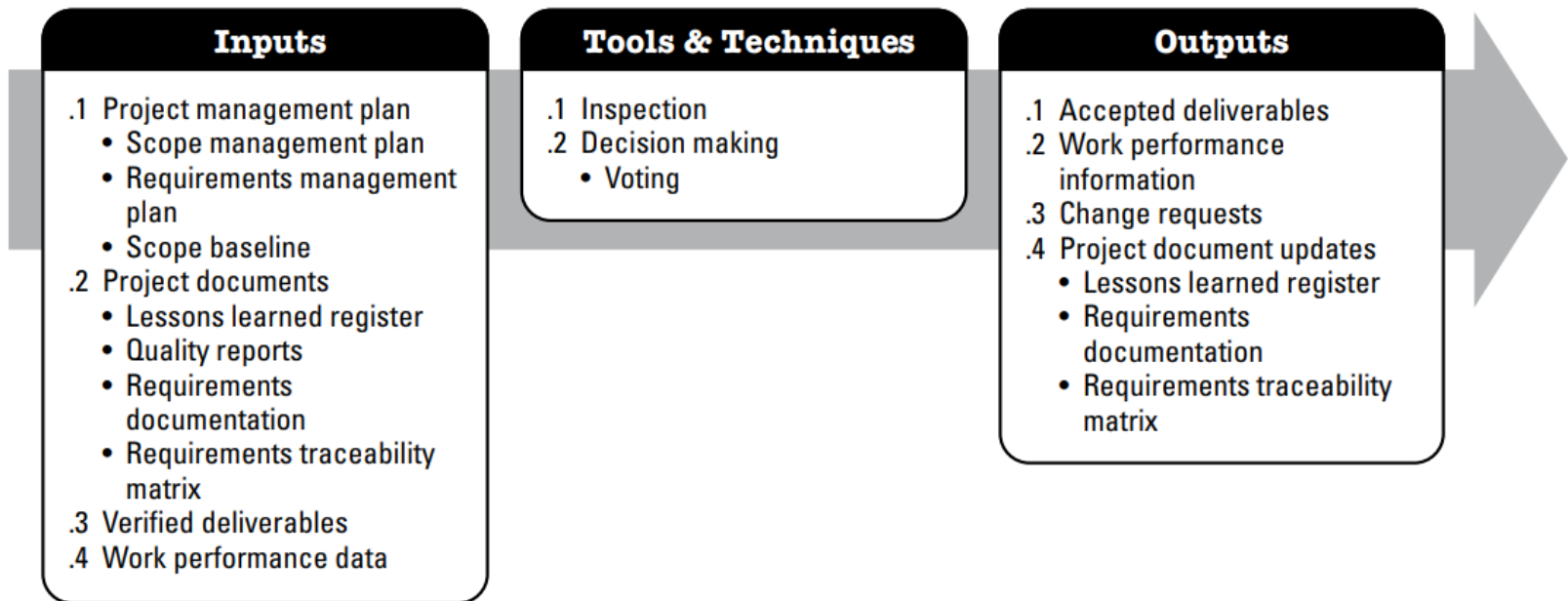
Validate Scope

Validate Scope

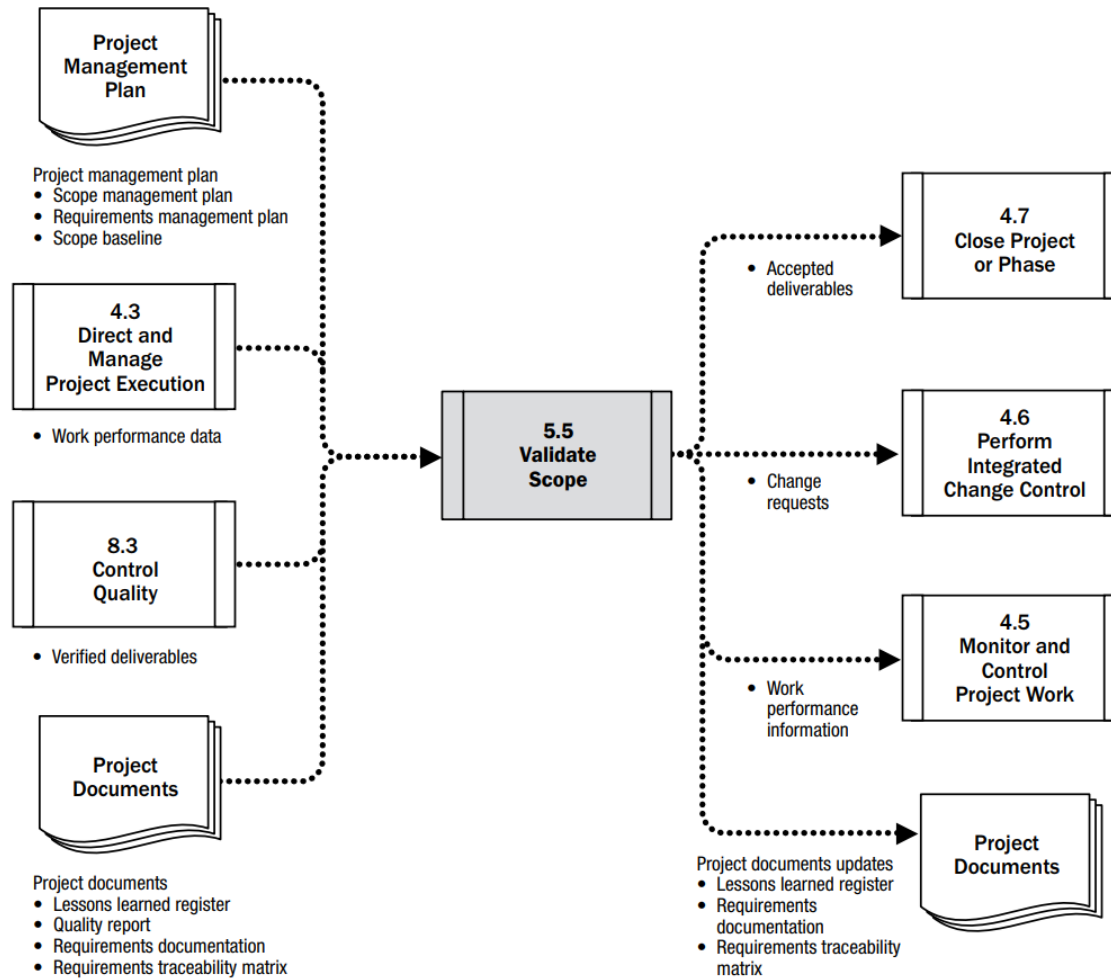
- **What:** process of formalizing acceptance of the completed project deliverables.
 - Actions to review and approve work packages and deliverables meet the acceptance criteria defined in the WBS dictionary.
- **Who:**
 - Customer/User
 - Project Sponsor
 - Authorized stakeholders
- **Why:** Decrease the reworks, increase likelihood of closing the project.
- **When:**
 - Periodically throughout the project as needed.
 - Usually performed after Control Quality, and before Close Project or Phase.



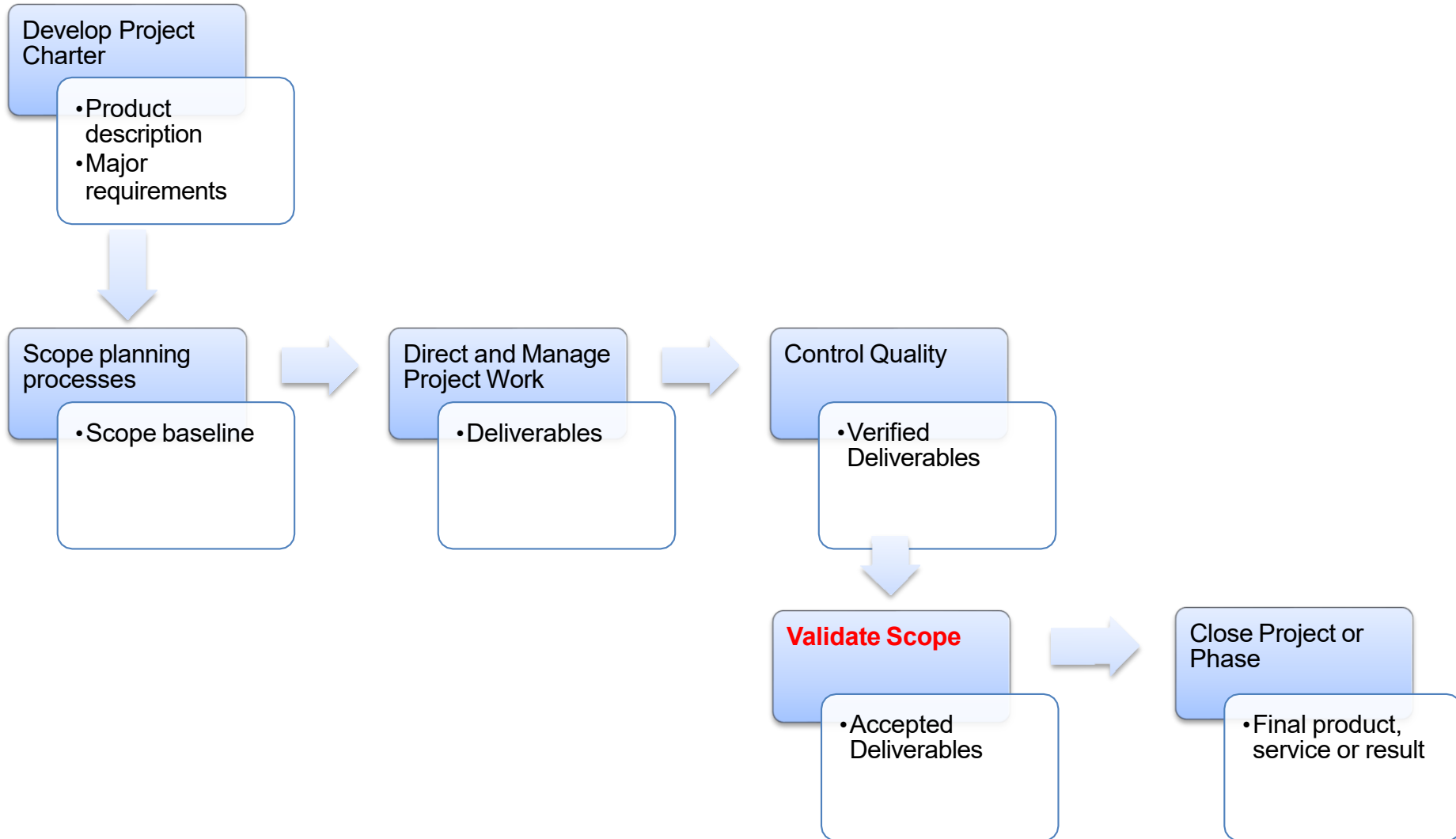
Process



Data Flow



Deliverables flow



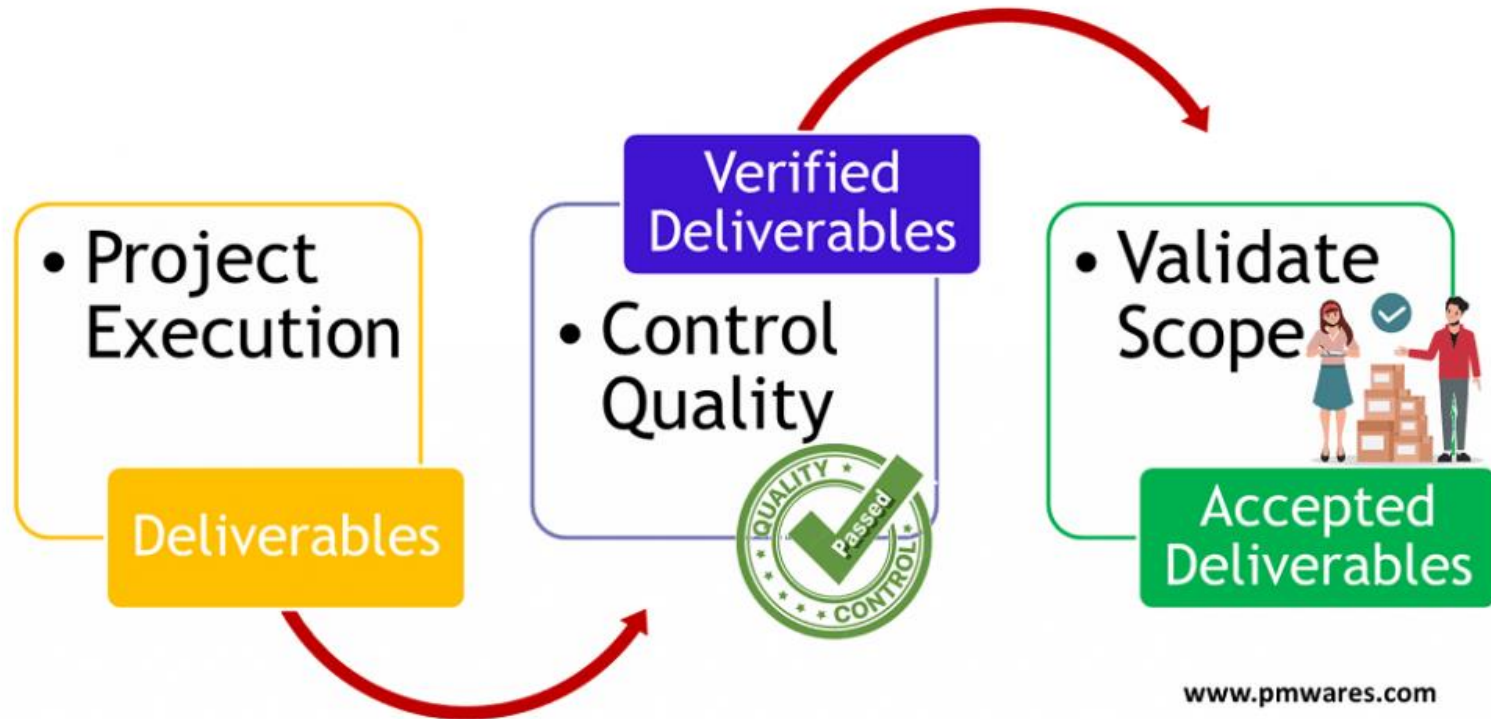
Validate scope

How?

- Project team presents project deliverables to customer or sponsor. After customer checks the deliverables, there are 3 possible outcomes:
- If the **deliverable meets their requirements** agreed in project scope, the customer accepts it (**formal acceptance**).
- If the deliverable has **defects** or requirements are not met, they must be fixed by the project team.
- If there are change request, they will be evaluated by the change control board (only approved change requests will be implemented in the project.)



Control Quality & Validate Scope



Control Quality & Validate Scope

Control Quality

- Process of ensuring that the deliverables **meet the quality requirements** defined in the Quality management plan.
- Focus on **Correctness of deliverables**.
- Performed by **Quality control department**.
- Belongs to **Quality Management** Knowledge Area.
- Output are **Verified deliverables**, and input to Control Quality process.
- Process performed **before Validate Scope process** or in parallel.

Validate Scope

- Process of **formalizing acceptance** of the completed project deliverables by customer.
- Focus on **Acceptance of deliverables**.
- Performed by **Customer or sponsor**.
- Belongs to **Scope Management** knowledge Area.
- Output are **Accepted deliverables**.
- Process performed **after Control Quality process** or in parallel.

Tools & Techniques

Inspection

- Sometimes called: reviews, product reviews, audits, and walkthroughs.
- A point-by-point review of the requirements and the associated deliverable/ work



Decision Making

- **Voting:** to reach a conclusion when the validation is performed by the project team and other stakeholders



Outputs

Accepted deliverables

- Deliverables that meet the acceptance criteria are **formally signed off** and approved by the customer or sponsor.



Work Performance Information

- Information about accepted deliverable and not-accepted deliverables and the reasons why.

Change requests

- Defect repair

Project documents updates

- Requirements documentation.
- Requirements traceability matrix.
- Lessons learned register.

Inputs

Verified deliverables

- Verified deliverables are project deliverables that are completed and checked for correctness through the Control Quality process.

Project Management Plan

- **Scope management plan**
- **Requirements management plan**
- **Scope baseline:** The scope baseline is compared to actual results to determine if a change

Project documents

- **Requirements Documentation:** requirements with their acceptance criteria
- **Requirements Traceability Matrix:**
 - links requirements to their origin
- **Quality Reports**
- **Lessons learned register**

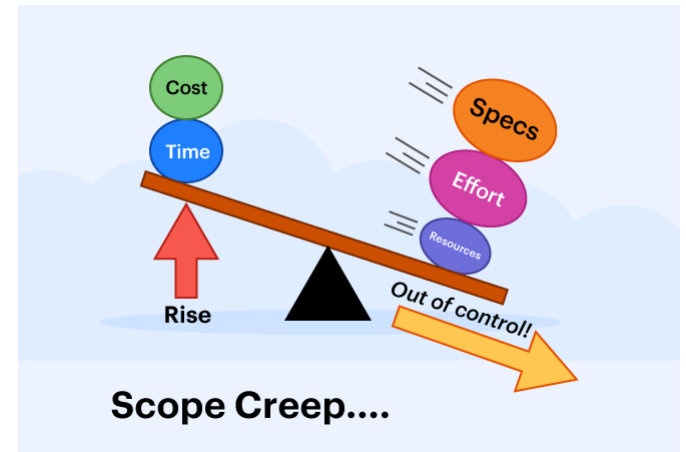
Work performance data

- the degree of compliance with requirements, number of nonconformities, severity of the nonconformities, or the number of validation cycles performed in a period of time.

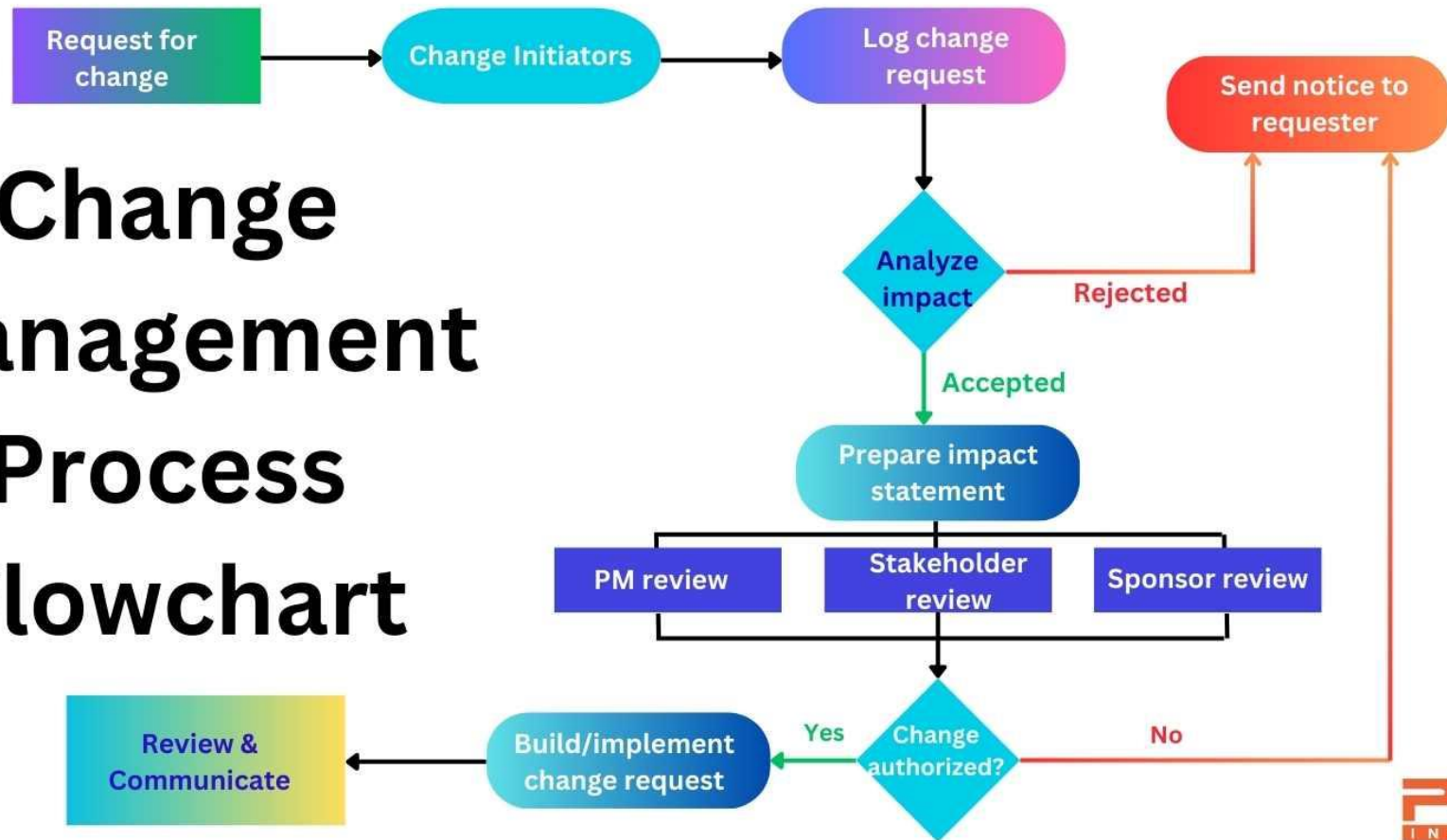
Control Scope

Control Scope

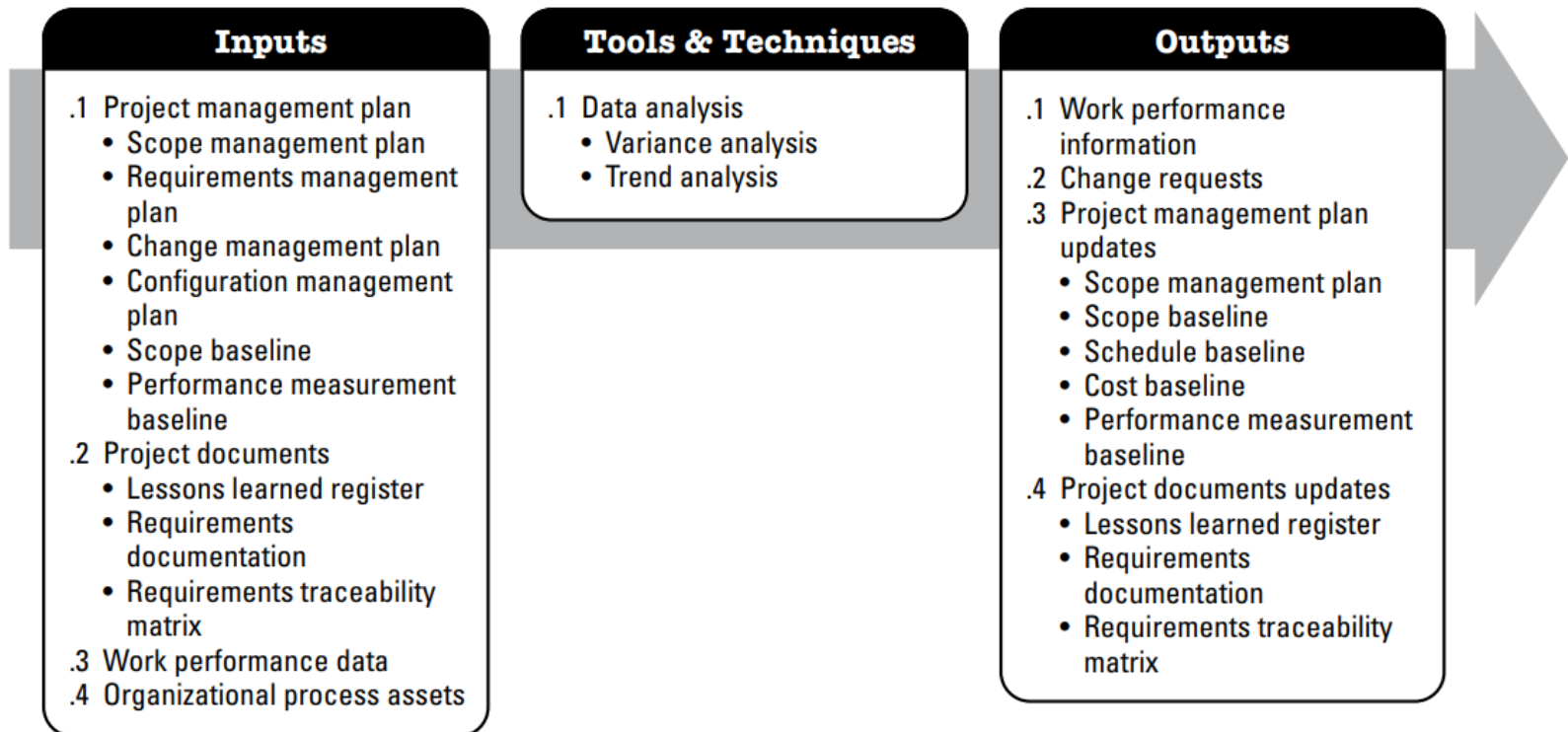
- **What:** process of monitoring the status of the project and product scope and managing changes to the scope baseline.
- **Why:**
 - Change is inevitable
 - The uncontrolled expansion to product or project scope without adjustments to time, cost, and resources is referred to as scope creep.
 - Ensures all requested changes and recommended corrective or preventive actions are processed through the Perform Integrated Change Control process
- **When:** Throughout the project.



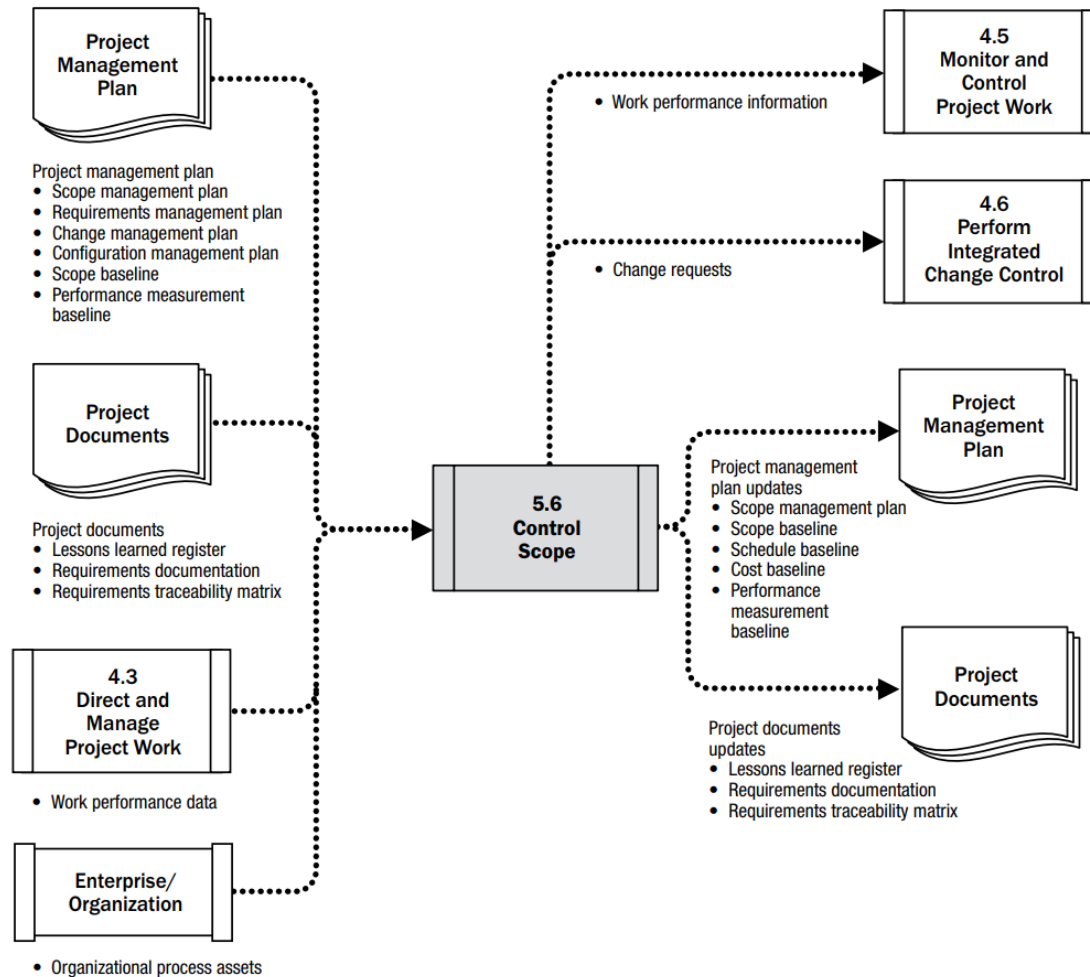
Change Management Process Flowchart



Process



Data Flow



Tools and Techniques

Variance Analysis

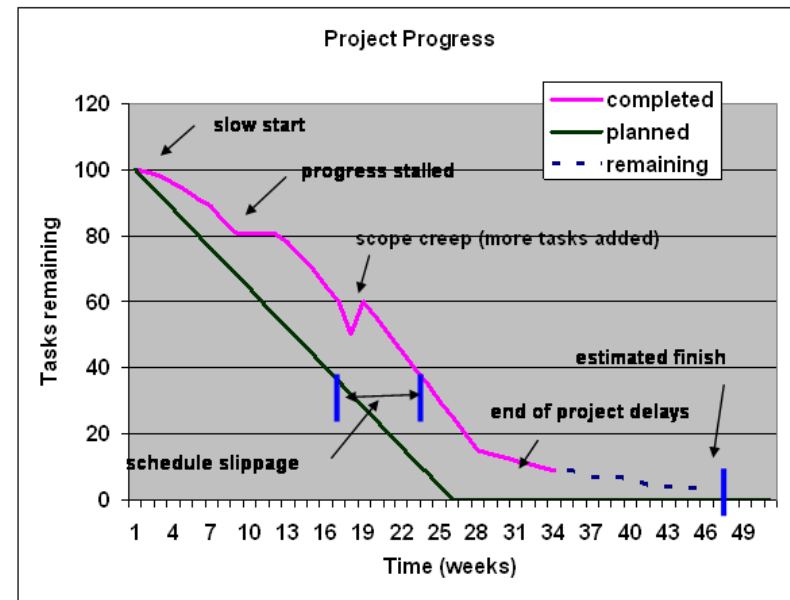
- Assess the magnitude of variation from the original scope baseline .
- Determine the cause and the degree of variance relative to the scope baseline
- Decide whether corrective or preventive action is required

Scope Variance Index (SVI):

$$SVI = \frac{\text{Actual Deliverables Completed}}{\text{Planned Deliverables}} = \frac{93}{127} \approx 0.73$$

Trend analysis

- examines project performance over time to determine if performance is improving or deteriorating.



Outputs

Work Performance Information

- **Planned vs. actual technical performance** or other scope performance measurements: the categories of the changes received, the identified scope variances and their causes, how they impact schedule or cost, and the forecast of the future scope performance.
- This information provides a foundation for making scope decisions.

Change requests

- Preventive, corrective actions or defect repairs.

Project management plan updates

- Scope management plan
- Scope baseline
- Other baselines (schedule, cost, performance measurement ..)

Project document updates

- Requirements documentation
- Requirements traceability matrix
- Lessons learned register

WBS ID	Task/Deliverable	Planned Work	Actual Work Done	Variance	Comments / Root Cause	Action Required
1.1	UI/UX Design	10 screens	8 screens	-2 screens	Delay due to feedback loop with client	Extend design by 2 days
1.2	Customer API Development	12 endpoints	12 endpoints	0	On track	None
1.3	Admin Panel	5 core modules	3 core modules	-2 modules	Dev resource reallocated to bug fixing	Add 1 developer temporarily
1.4	Testing Preparation	100 test cases	70 test cases	-30 test cases	Late requirements freeze; test case drafting delayed	Revise QA resource allocation

Project Progress

